

36. EtherCAT Slave Controller (ESC)

36.1 Overview

The EtherCAT slave controller (ESC) uses an EtherCAT Slave Controller IP Core made by Beckhoff Automation GmbH, Germany. The ESC handles EtherCAT communications as an interface between the EtherCAT field bus and slave applications.

Table 36.1 Specification of the EtherCAT slave controller

Item		Description
Number of ports		2
FMMU		8
SyncManager		8
Process data RAM [KB]		8
Distributed clocks		64 bits
EBus		No
Process data interfaces (PDI)	Digital I/O	No
	SPI slave	No
	Host MPU interface	On-chip bus

Figure 36.1 shows a block diagram of the EtherCAT slave controller.

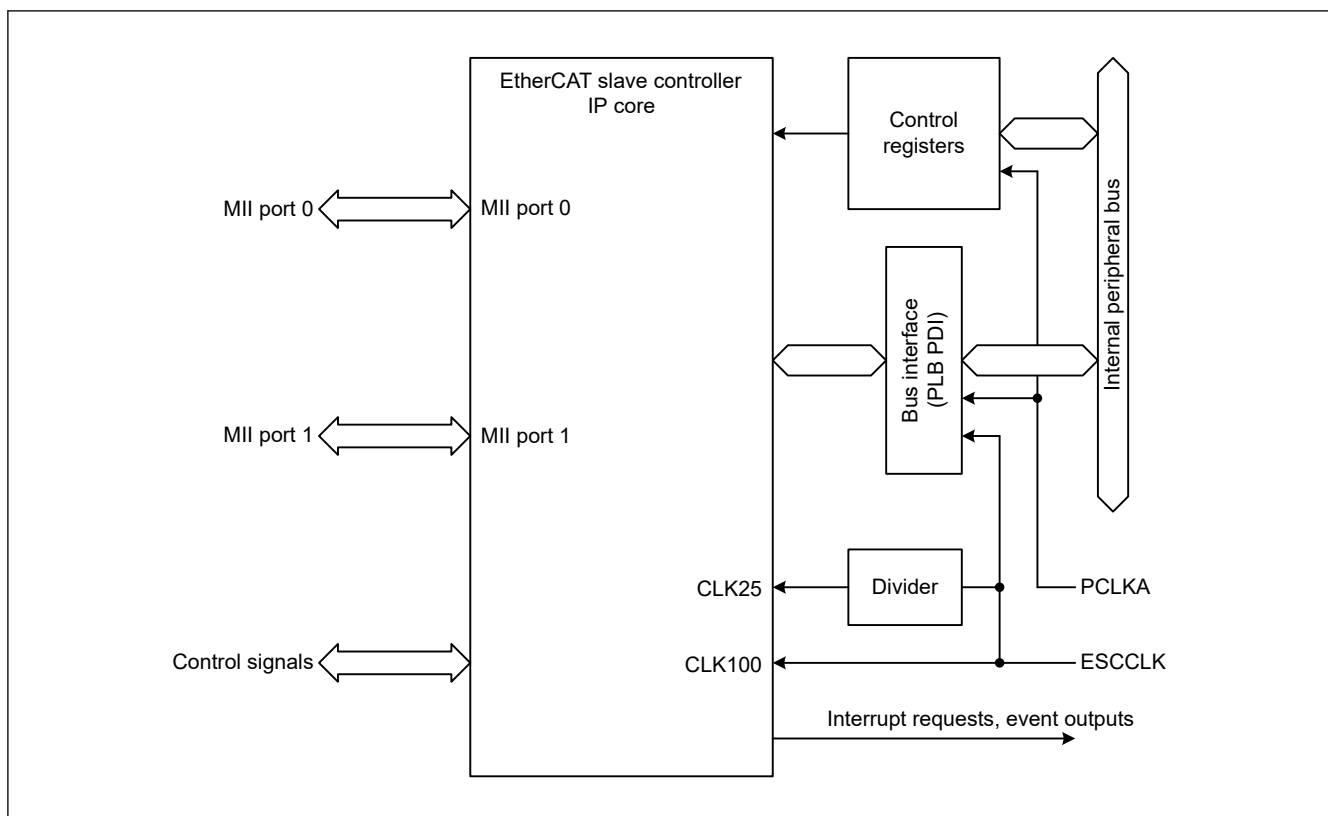


Figure 36.1 Block diagram of the EtherCAT slave controller

Table 36.2 lists the input/output pins of the EtherCAT controller.

Table 36.2 Input/Output pins of the EtherCAT slave controller
(excluding PHY MII pins)

Pin Name	I/O	Description
CATn_LINKSTA	Input	Link status inputs from the PHY-LSI
CATn_RX_CLK	Input	Receive clocks
CATn_RX_DV	Input	Receive data valid
CATn_ERXD3 to CATn_ERXD0	Input	4-bit receive data
CATn_RX_ER	Input	Receive error
CATn_TX_CLK	Input	Transmit clocks
CATn_TX_EN	Output	Transmit enable
CATn_ETXD3 to CATn_ETXD0	Output	4-bit transmit data
CAT0_MDC	Output	Management data clock
CAT0_MDIO	I/O	Management data I/O
ETHPHYCLK	Output	25 MHz clock signal
CATRESTOUT	Output	PHY reset signal
CATLINKACTn	Output	Link/Activity LED outputs
CATLEDRUN	Output	RUN LED (green) output
CATLEDERR	Output	ERR LED (red) output
CATLEDSTER	Output	Output for RUN LED part of STATE LED (bicolor) (turned off while ERR)
CATI2CCLK	Output	EEPROM I2C clock output
CATI2CDATA	I/O	EEPROM I2C data
CATSYNCn	Output	SYNC signal outputs
CATLATCHn	Input	LATCH signal inputs
CATIRQ	Output	IRQ output

Figure 36.2 shows an example of connecting the MCU to the PHY-LSI.

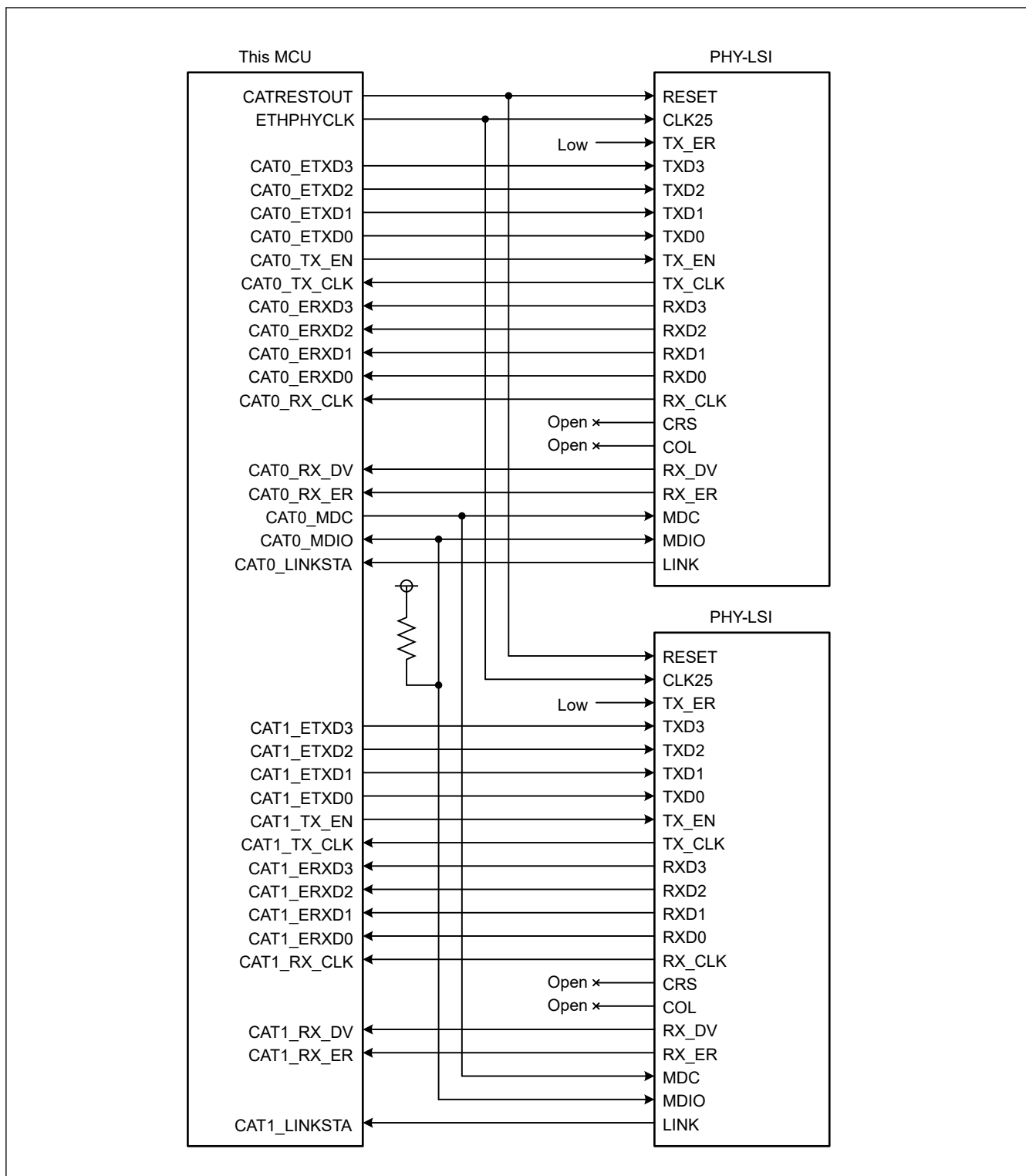


Figure 36.2 Example of Connection with the PHY-LSI

36.2 Functional Overview

Typical functions of the EtherCAT slave controller and supported functions by this LSI are shown in [Table 36.3](#). Regarding the detailed specification of EtherCAT and ESC, refer to the documentation (ETG.1000 EtherCAT Specification) provided by EtherCAT Technology Group (ETG) and the EtherCAT Slave Controller IP Core (v2.04) data sheet provided by Beckhoff Automation.

Table 36.3 Typical functions of EtherCAT slave controller and supported functions by this LSI (1 of 3)

Features	Functions	Support
EtherCAT protocol	Handling the following frames: • Ethernet frames with Ether Type of 0x88A4 • EtherCAT frames encapsulated in UDP/IP • EtherCAT frames with VLAN Tag • Normal Ethernet frames	✓
Addressing modes	Device addressing • Auto increment address • Configured station address • Broadcast address	✓
	Logical addressing	✓
Working counter	Counting the number of read/write from/to the device	✓
EtherCAT command type	Processing the command that master requests slaves to address each addressing mode	✓
Loop control	Loop control and loop state in ESC	✓
Shadow buffer	Shadow buffers function when register is read/written	✓
Circulating frames	Processing of circulating frames during the failure	✓
Link detection	LINK_MII signal (PHY link signal)	✓
	MI Link detection and configuration (monitoring the PHY register via the management interface)	—
	Enhanced link detection (monitoring the state of transfer by MII receive error monitor)	✓
FIFO size reduction	Receive FIFO size reduction for reduction of propagation delay	✓
Ethernet physical layer	MII	✓
	EBUS	—
	Back-to-Back MII connection	✓
	MII management interface	✓
	Read/write of the PHY register via MII management interface	✓
	PHY address offset	✓
	Manual TX shift compensation	✓
	Automatic TX shift compensation	✓
FMMU	Mapping between logical addresses and physical addresses	✓
SyncManager	Buffered mode	✓
	Mailbox mode	✓
	Interrupt and latch event generation when a buffer was completely and successfully written or read.	✓
	Repeating mailbox communication	✓
	SyncManager deactivation by the PDI	✓

Table 36.3 Typical functions of EtherCAT slave controller and supported functions by this LSI (2 of 3)

Features	Functions	Support
Distributed clocks	Clock Synchronization considering propagation delay and drift compensation	✓
	Generation of SyncSignals (SYNC0 and SYNC1 output signals) • Cyclic generation mode • Single shot mode • Cyclic acknowledge mode • Single shot acknowledge mode	✓
	Precise time stamping of LatchSignal events (LATCH0 and LATCH1 input signals) • Single event mode • Continuous mode • SyncManager event mode (for debugging)	✓
	Generation of synchronous interrupts	✓
	Synchronous digital output updates / Synchronous digital input sampling	—
	Exclusive access to the Sync unit and Latch units by the ECAT and PDI	✓
	System time PDI controlled	—
	Communication Timing • Free run • Synchronized to output event • Synchronized to SyncSignal	✓
EtherCAT state machine	Control of state machine / Indication of the states and error code	✓
	Device emulation	—
SII EEPROM	SII EEPROM commands	✓
	SII EEPROM error indication	✓
	SII EEPROM access interface	✓
	EEPROM size configuration	✓
	EEPROM emulation	—
Interrupts	AL event request (PDI interrupt)	✓
	ECAT event request (ECAT interrupt)	✓
Watchdogs	Process data watchdog	✓
	PDI watchdog	✓
Error counters	Port error counters	✓
	Forwarded RX error counter	✓
	ECAT processing unit error counter	✓
	PDI error counter	✓
	Lost link counter	✓
	Process data watchdog counter	✓
	PDI watchdog counter	✓
LED signals	RUN LED signal	✓
	ERR LED signal	✓
	STATE LED support and STATE_RUN LED signal	✓
	LINKACT LED signal	✓
	Port error LED signal	—
	RUN/ERR LED override	✓

Table 36.3 Typical functions of EtherCAT slave controller and supported functions by this LSI (3 of 3)

Features	Functions	Support
Process data interface (PDI)	Digital I/O	—
	SPI slave	—
	8-bit/16-bit synchronous/asynchronous microcontroller interface	—
	On-chip bus	✓
	General purpose I/O	—
Write protection	Write protection for the register area (0x0000 to 0x0F7F)	✓
	Write protection for the whole area including the user RAM and process data RAM (0x0000 to 0x2FFF)	✓
ESC reset	ESC reset by the EtherCAT master or via the PDI	✓

36.3 Description of Registers

The ESC has two types of registers: registers in the EtherCAT slave controller IP Core (ESC IP Core) itself and registers to generate signals for controlling the ESC IP Core. Those of the former type are assigned to addresses from 0x430A_0000 to 0x430A_0FFF and those of the latter type are assigned to addresses 0x403A_4000 to 0x403A_4023.

All the bit function tables of the registers in the ESC IP Core have two separate R/W accessibility columns, one indicating the case for the PDI and the other that for the ECAT. The PDI column indicates accessibility from the CPU, and the ECAT column indicates accessibility from the EtherCAT master.

36.3.1 ESCRST : ESC Reset Control Register

Base address: ESC_INI = 0x403A_4000
ESC_INI_NS = 0x503A_4000

Offset address: 0x000

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	PHYRST	ESCRST
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ESCRST	ESC Reset 0: Reset 1: Reset is released	R/W
1	PHYRST	PHY Reset 0: Reset 1: Reset is released	R/W
31:2	—	The read values are undefined. The write value should be 0.	R/W

This register is used to reset the ESC IP Core and PHY-LSIs.

ESCRST bit (ESC Reset)

Setting this bit to 0 resets all registers in the ESC IP Core.

This bit cannot be used to initiate driving of the CATRESTOUT pin low because the reset output from the ESC IP Core becomes 0 at this time.

When a reset is requested (when the ESC issues a RESET interrupt request) by ECAT or PDI, set this bit to 0.

ESC registers cannot be accessed immediately after ESCRST.ESCRST bit is set to 1. When ESCRST.ESCRST bit is set to 1, wait time is needed at least 4 cycles in PCLKA before the next register access.

PHYRST bit (PHY Reset)

Setting this bit to 0 drives the CATRESTOUT pin low.

When the ESCRST bit is to be set to 0, set this bit to 0 beforehand.

36.3.2 PHYLINK : PHY LINK Polarity Setting Register

Base address: ESC_INI = 0x403A_4000
ESC_INI_NS = 0x503A_4000

Offset address: 0x010

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	LINKP OL1	LINKP OL0
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	LINKPOL0	CAT0_LINKSTA Pin Polarity 0: Active high 1: Active Low	R/W
1	LINKPOL1	CAT1_LINKSTA Pin Polarity 0: Active high 1: Active Low	R/W
31:2	—	The read values are undefined. The write value should be 0.	R/W

This register is used to set the signal polarity of CATn_LINKSTA pins (n = 0, 1).

36.3.3 ESCICR : ESC Interrupt Control Register

Base address: ESC_INI = 0x403A_4000
ESC_INI_NS = 0x503A_4000

Offset address: 0x014

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	LAT1S	LAT0S	—	ESCIC	SYNC 1C	SYNC 0C
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	SYNC0C	SYNC0 Interrupt Generating Condition Setting Condition Setting 0: An interrupt is generated at the rising edge of the SYNC0 signal. 1: An interrupt is generated at the falling edge of the SYNC0 signal.	R/W
1	SYNC1C	SYNC1 Interrupt Generating Condition Setting Condition Setting 0: An interrupt is generated at the rising edge of the SYNC1 signal. 1: An interrupt is generated at the falling edge of the SYNC1 signal.	R/W

Bit	Symbol	Function	R/W
2	ESCIC	ESCI Interrupt Generating Condition Setting PHY 0: An interrupt is generated when the PDI_IRQ is 1. 1: An interrupt is generated when the PDI_IRQ is 0.	R/W
3	—	The read value is undefined. The write value should be 0.	R/W
4	LAT0S	Select DC Latch Trigger 0 for ESC 0: Select LATCH0 input from external 1: Select LATCH0 input from ELC	R/W
5	LAT1S	Select DC Latch Trigger 1 for ESC 0: Select LATCH1 input from external 1: Select LATCH1 input from ELC	R/W
31:6	—	The read values are undefined. The write value should be 0.	R/W

This register is used to set the conditions for the generation of interrupts by the ESC.

36.3.4 ECATOFFADR : EtherCAT PHY Offset Address Register

Base address: ESC_INI = 0x403A_4000
ESC_INI_NS = 0x503A_4000

Offset address: 0x018

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	OADD[4:0]				
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
4:0	OADD[4:0]	PHY Offset Address Setting Set the offset address of PHY of using the EtherCAT.	R/W
31:5	—	The read values are undefined. The write value should be 0.	R/W

This register is used to set the conditions for the generation of interrupts by the ESC..

OADD[4:0] bit (PHY Offset Address Setting)

These bits are used to offset the addresses specified in the PHYAD fields of MII management frames.

Typically, usage is with the bits set to 00000b. In this case, the port numbers match the PHY addresses.

In cases where address 0 is not available for use, such as with some PHY-LSIs in which a broadcast address is associated with address 0, use these bits to offset the addresses.

36.3.5 ECATOPMOD : EtherCAT Operation Mode Register

Base address: ESC_INI = 0x403A_4000
ESC_INI_NS = 0x503A_4000

Offset address: 0x01C

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	EEPR OMSIZ E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
0	EEPROMSIZE	Sets the size of the EEPROM memory of the EtherCAT. 0: 16 Kbits or less 1: 32 Kbits to 4 Mbits	R/W
31:1	—	The read values are undefined. The write value should be 0.	R/W

EEPROMSIZE bit (Sets the size of the EEPROM memory of the EtherCAT.)

This bit is used to set the memory size of the SII EEPROM to be used.

36.3.6 ECATDBGC : EtherCAT Debug Control Register

Base address: ESC_INI = 0x403A_4000
ESC_INI_NS = 0x503A_4000

Offset address: 0x0020

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	TXSFT1[1:0]	TXSFT0[1:0]	—	—
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
1:0	TXSFT0[1:0]	Port 0 TX Signal Delay Setting 0 0: 0 ns 0 1: 10 ns 1 0: 20 ns 1 1: 30 ns	R/W
3:2	TXSFT1[1:0]	Port 1 TX Signal Delay Setting 0 0: 0 ns 0 1: 10 ns 1 0: 20 ns 1 1: 30 ns	R/W
31:4	—	The read values are undefined. The write value should be 0.	R/W

TXSFTn[1:0] Bits (Port n TX Signal Delay Setting) (n = 0, 1)

These bits are used to set the amount of delay on the CATn_TX_EN and CATn_ETXD0 to CATn_ETXD3 signals of the respective ports when manual TX shift compensation is used. Select the amount of delay so that the setup and hold requirements relative to the CATn_TX_CLK signals are satisfied.

To use this function, set the CATn_TX_CLK pin to the low level.

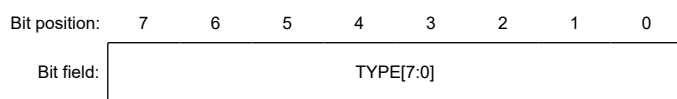
36.4 ESC Information Register

In the tables, ECAT indicates access by the EtherCAT master controller and PDI indicates access by the local CPU.

36.4.1 TYPE : Type Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0000



Value after reset: 1 0 1 0 0 0 0 0

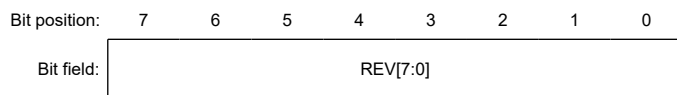
Bit	Symbol	Function	R/W
7:0	TYPE[7:0]	Type of the EtherCAT slave controller	R

The TYPE register indicates the type of the EtherCAT slave controller.

36.4.2 REVISION : Revision Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0001



Value after reset: 0 0 0 0 0 0 1 0

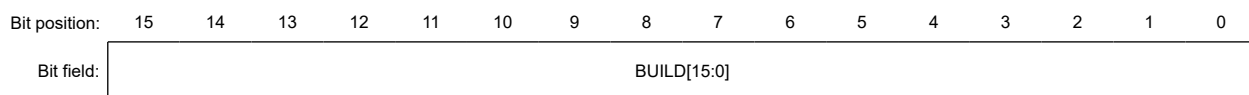
Bit	Symbol	Function	R/W
7:0	REV[7:0]	Revision of the EtherCAT slave controller	R

The REVISION register indicates the revision of the EtherCAT slave controller.

36.4.3 BUILD : Build Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0002



Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	BUILD[15:0]	Build number of the EtherCAT slave controller	R

The BUILD register indicates the build number of the EtherCAT slave controller.

36.4.4 FMMU_NUM : FMMU Supported Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0004

Bit position: 7 6 5 4 3 2 1 0

Bit field: NUMFMMU[7:0]

Value after reset: 0 0 0 0 1 0 0 0

Bit	Symbol	Function	R/W
7:0	NUMFMMU[7:0]	Number of FMMU channels supported in the EtherCAT slave controller	R

The FMMU_NUM register indicates the number of FMMU channels supported in the EtherCAT slave controller.

36.4.5 SYNC_MANAGER : SyncManager Supported Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0005

Bit position: 7 6 5 4 3 2 1 0

Bit field: NUMSYNC[7:0]

Value after reset: 0 0 0 0 1 0 0 0

Bit	Symbol	Function	R/W
7:0	NUMSYNC[7:0]	Number of SyncManager channels supported in the EtherCAT slave controller	R

The SYNC_MANAGER register indicates the number of SyncManager channels supported in the EtherCAT slave controller.

36.4.6 RAM_SIZE : RAM Size Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0006

Bit position: 7 6 5 4 3 2 1 0

Bit field: RAMSIZE[7:0]

Value after reset: 0 0 0 0 1 0 0 0

Bit	Symbol	Function	R/W
7:0	RAMSIZE[7:0]	Process data RAM size supported in the EtherCAT slave controller (unit: KB)	R

The RAM_SIZE register indicates the process data RAM size supported in the EtherCAT slave controller in KB.

36.4.7 PORT_DESC : Port Descriptor Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0007

Bit position: 7 6 5 4 3 2 1 0

Bit field:	P3[1:0]	P2[1:0]	P1[1:0]	P0[1:0]
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Value after reset: 0 0 1 1 1 1 1 1

Bit	Symbol	Function	R/W
1:0	P0[1:0]	Port 0 configuration Fixed to the setting for MII connection (11b) in this LSI. 0 0: Not implemented 0 1: Not configured (SII EEPROM) 1 0: EBUS 1 1: MII	R
3:2	P1[1:0]	Port 1 configuration Fixed to the setting for MII connection (11b) in this LSI. 0 0: Not implemented 0 1: Not configured (SII EEPROM) 1 0: EBUS 1 1: MII	R
5:4	P2[1:0]	Port 2 configuration Fixed to the setting for MII connection (11b) in this LSI. 0 0: Not implemented 0 1: Not configured (SII EEPROM) 1 0: EBUS 1 1: MII	R
7:6	P3[1:0]	Port 3 configuration This LSI does not implement port 3. Fixed to 00b. 0 0: Not implemented 0 1: Not configured (SII EEPROM) 1 0: EBUS 1 1: MII	R

The PORT_DESC register indicates the port configuration.

36.4.8 FEATURE : ESC Features Supported Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0008

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	FSCO NFIG	RWSU PP	LRW	DCSY NC	FCS	LINKD ECMII	—	—	DCWI D	DC	—	FMMU
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Value after reset: x x x x 0 0 0 1 1 1 x x 1 1 x 0

Bit	Symbol	Function	R/W
0	FMMU	FMMU Operation 0: Bit oriented 1: Byte oriented	R
1	—	The read value is undefined.	R
2	DC	Distributed Clock 0: Not available 1: Available	R

Bit	Symbol	Function	R/W
3	DCWID	Distributed Clock Width 0: 32 bits 1: 64 bits	R
5:4	—	The read values are undefined.	R
6	LINKDECMII	Enhanced Link Detection in MII 0: Not available 1: Available	R
7	FCS	Separate handling of FCS errors 0: Not supported 1: Supported. Frames with wrong FCS and additional nibble will be counted separately in forwarded RX error counter.	R
8	DCSYNC	Enhanced DC SYNC activation 0: Not available 1: Available	R
9	LRW	EtherCAT LRW command support 0: Supported 1: Not supported	R
10	RWSUPP	EtherCAT read/write command support (BRW, APRW, FPRW) 0: Supported 1: Not supported	R
11	FSCONFIG	Fixed FMMU/SyncManager configuration 0: Variable configuration 1: Fixed configuration	R
15:12	—	The read values are undefined.	R

The FEATURE register indicates the features supported in the EtherCAT slave controller.

36.5 Station Address Registers

36.5.1 STATION_ADR : Configured Station Address Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0010

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: NODADDR[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	NODADDR[15:0]	Node Addressing Address Indication Address used for node addressing (FPxx commands) The access type from the EtherCAT master is R/W.	R

The STATION_ADR register indicates the address used for node addressing.

36.5.2 STATION_ALIAS : Configured Station Alias Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0012

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	NODALIADDR[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	NODALIADDR[15:0]	Alias Address Indication Alias address used for node addressing (FPxx commands). The use of this alias is activated by setting bit 24 of the ESC DL control register (ESC_DL_CONTROL at 0x0100) to 1. The access type from the EtherCAT master is R.	R/W

Note: The initial value, 0, is retained until the EEPROM is loaded. After that, the value becomes the value at address 0x0004 in the EEPROM. This value is only taken over from the EEPROM the first time the EEPROM is loaded after a power-on or reset.

The STATION_ALIAS register indicates the alias address used for node addressing (FPxx commands).

36.6 Write Protection Registers

36.6.1 WR_REG_ENABLE : Write Register Enable Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0020

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ENAB LE
Value after reset:	x	x	x	x	x	x	x	0

Bit	Symbol	Function	R/W
0	ENABLE	Register Write Protection Unlock When registers are currently being protected against writing (bit 0 is 1 in the write register protection register, WR_REG_PROTECT, at 0x0021) and freely writing to registers of the given node is to be permitted, the operation to do so by writing to this register has to proceed in the same Ethernet frame and preceding the other desired writing to registers. Write protection will be reactivated once the frame period elapses (unless the value in the write register protection register is changed). The access type from the EtherCAT master is R/W.	R
7:1	—	The read values are undefined.	R

The WR_REG_ENABLE register is used to unlock the write protection temporarily while registers are write-protected.

36.6.2 WR_REG_PROTECT : Write Register Protection Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0021

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PROT ECT
Value after reset:	x	x	x	x	x	x	x	0

Bit	Symbol	Function	R/W
0	PROTECT	Register Write Protection Specification The access type from the EtherCAT master is R/W. 0: Protection disabled 1: Protection enabled	R
7:1	—	The read values are undefined.	R

The WR_REG_PROTECT register is used to protect registers against writing. The registers in the area 0x8013_0000 to 0x8013_0FFF are write-protected (except for the WR_REG_ENABLE register (0x0020) and ESC_WR_ENABLE register (0x0030)).

36.6.3 ESC_WR_ENABLE : ESC Write Enable Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0030

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ENAB LE

Value after reset: x x x x x x x 0

Bit	Symbol	Function	R/W
0	ENABLE	Register/Memory Write Protection Unlock When registers are currently being protected against writing by ESC write protection (bit 0 is 1 in the ESC write protection register, ESC_WR_PROTECT, at 0x0031) and freely writing to registers of the given node is to be permitted, the operation to do so by writing to this register has to proceed in the same Ethernet frame and preceding the other desired writing to registers. Write protection will be reactivated once the frame period elapses (unless the value in the ESC write protection register is changed). The access type from the EtherCAT master is R/W.	R
7:1	—	The read values are undefined.	R

The ESC_WR_ENABLE register is used to unlock the write protection temporarily while registers and memories are write-protected by ESC write protection.

36.6.4 ESC_WR_PROTECT : ESC Write Protection Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0031

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PROT ECT

Value after reset: x x x x x x x 0

Bit	Symbol	Function	R/W
0	PROTECT	Register/Memory Write Protection Specification The access type from the EtherCAT master is R/W. 0: Protection disabled 1: Protection enabled	R
7:1	—	The read values are undefined.	R

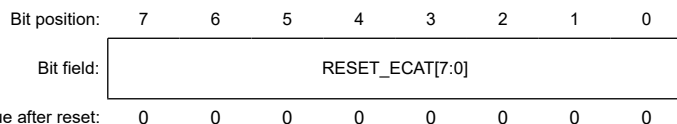
The ESC_WR_PROTECT register is used to protect registers against writing. Registers and memories in the area 0x8013_0000 to 0x8013_2FFF including the process data RAM are write-protected (except for the WR_REG_ENABLE register (0x0020) and ESC_WR_ENABLE register (0x0030)).

36.7 Data Link Layer Registers

36.7.1 ESC_RESET_ECAT_W : ESC Reset ECAT Register for write

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0040



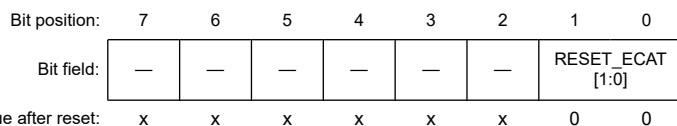
Bit	Symbol	Function	R/W
7:0	RESET_ECAT[7:0]	Software Reset Setting A reset is enabled after writing 0x52 ('R'), 0x45 ('E'), and 0x53 ('S') consecutively to this register. The access type from the EtherCAT master is R/W.	R

This register is used to reset the EtherCAT slave controller from the ECAT (master) by software.

36.7.2 ESC_RESET_ECAT_R : ESC Reset ECAT Register for read

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0040



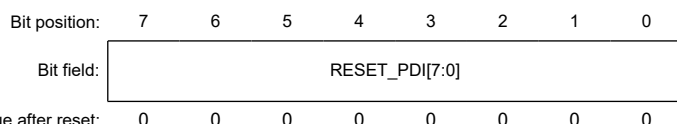
Bit	Symbol	Function	R/W
1:0	RESET_ECAT[1:0]	Reset Progress Status The access type from the EtherCAT master is R/W. 0 1: After writing 0x52 1 0: After writing 0x45 (if 0x52 was written before) 0 0: Others	R
7:2	—	The read values are undefined. The access type from the EtherCAT master is R/W. In this case, the write value should be 0.	R

This register is used to reset the EtherCAT slave controller from the ECAT (master) by software.

36.7.3 ESC_RESET_PDI_W : ESC Reset PDI Register for write

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0041



Bit	Symbol	Function	R/W
7:0	RESET_PDI[7:0]	Software Reset Setting A reset is enabled after writing 0x52 ('R'), 0x45 ('E'), and 0x53 ('S') consecutively to this register. The access type from the EtherCAT master is R.	R/W

This register is used to reset the EtherCAT slave controller from the PDI (slave) by software.

36.7.4 ESC_RESET_PDI_R : ESC Reset PDI Register for read

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0041

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	—	—	—	RESET_PDI [1:0]
------------	---	---	---	---	---	---	--------------------

Value after reset: x x x x x x 0 0

Bit	Symbol	Function	R/W
1:0	RESET_PDI[1:0]	Reset Progress Status The access type from the EtherCAT master is R. 0 1: After writing 0x52 1 0: After writing 0x45 (if 0x52 was written before) 0 0: Others	R/W
7:2	—	The read values are undefined. The write value should be 0.	R/W

This register is used to reset the EtherCAT slave controller from the PDI (slave) by software.

36.7.5 ESC_DL_CONTROL : ESC DL Control Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0100

Bit position: 31 30 29 28 27 26 25 24 23 22 21 20 19 18 17 16

Bit field:	—	—	—	—	—	—	—	STAAL IAS	—	—	—	—	—	RXFIFO[2:0]
------------	---	---	---	---	---	---	---	--------------	---	---	---	---	---	-------------

Value after reset: x x x x x x x 0 x x x x x 1 1 1

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:	LP3[1:0]	LP2[1:0]	LP1[1:0]	LP0[1:0]	—	—	—	—	—	—	TEMP USE	FWDR ULE
------------	----------	----------	----------	----------	---	---	---	---	---	---	-------------	-------------

Value after reset: 1 1 0 0 0 0 0 0 x x x x 0 0 0 1

Bit	Symbol	Function	R/W
0	FWDRULE	Forwarding Rule The source MAC address is changed for every frame (SOURCE_MAC[1] is set to 1 (locally administered address)) regardless of the forwarding rule. The access type from the EtherCAT master is R/W. 0: EtherCAT frames are processed. Non-EtherCAT frames are forwarded without processing. 1: EtherCAT frames are processed. Non-EtherCAT frames are destroyed.	R
1	TEMPUSE	Temporary Use of Bits 15 to 8 Settings The access type from the EtherCAT master is R/W. 0: Permanent use 1: Use for about 1 second, then revert to previous settings	R
3:2	—	These bits are read as 0.	R

Bit	Symbol	Function	R/W
7:4	—	The read values are undefined.	R
9:8	LP0[1:0]	Loop Port 0 Configuration The access type from the EtherCAT master is R/W. 0 0: Auto 0 1: Auto close 1 0: Open 1 1: Closed	R
11:10	LP1[1:0]	Loop Port 1 Configuration The access type from the EtherCAT master is R/W. 0 0: Auto 0 1: Auto close 1 0: Open 1 1: Closed	R
13:12	LP2[1:0]	Loop Port 2 Configuration Port 2 is not available on this LSI. The access type from the EtherCAT master is R/W. 0 0: Auto 0 1: Auto close 1 0: Open 1 1: Closed	R
15:14	LP3[1:0]	Loop Port 3 Configuration Port 3 is not available on this LSI. The access type from the EtherCAT master is R/W. 0 0: Auto 0 1: Auto close 1 0: Open 1 1: Closed	R
18:16	RXFIFO[2:0]	RX FIFO Size Set the RX FIFO size. The transfer time can be reduced by reducing the FIFO size. The access type from the EtherCAT master is R/W. 0 x x: Shortened by 40 ns 1 1 1: Default Others: No change	R
23:19	—	The read values are undefined.	R
24	STAALIAS	Station Alias Status The access type from the EtherCAT master is R/W. 0: Ignore station alias 1: Alias can be used for all configured address command types such as FPRD, FPWR.	R
31:25	—	The read values are undefined.	R

Note:

- Changes to loop configurations are delayed until any current reception or transmission of a frame through the port is completed.
- Reducing the size of the RX FIFO depends on all masters and slaves connected to the same network as the EtherCAT having very precise clock sources. An RX FIFO size of 7 (default) is sufficient if the precision of all clocks is 100 ppm or better and RX FIFO sizes of 0 to 3 are possible if the precision is 25 ppm or better.

The ESC_DL_CONTROL register is used to control loop in the EtherCAT slave controller and configure the RX FIFO size and station alias.

36.7.6 PHYSICAL_RW_OFFSET : Physical Read/Write Offset Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0108

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: RWOFFSET[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	RWOFFSET[15:0]	Offset between Read and Write Addresses Offset of R/W commands (FPRW, APRW) between read address and write address. When reading, RD_ADR = ADR (the given address is read) When writing, WR_ADR = ADR + R/W offset (writing is to the address obtained by adding the offset set in this register to the given address) The access type from the EtherCAT master is R/W.	R

The PHYSICAL_RW_OFFSET register is used to set the offset between read address and write address in the R/W commands.

36.7.7 ESC_DL_STATUS : ESC DL Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0110

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	COMP 3	LP3	COMP 2	LP2	COMP 1	LP1	COMP 0	LP0	PHYP 3	PHYP 2	PHYP 1	PHYP 0	—	ENHLI NKD	PDIW DST	PDIOP E
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	x	1	0	0

Bit	Symbol	Function	R/W
0	PDIOP E	PDI/EEPROM Load State Indication 0: EEPROM not loaded, the PDI not operational (process data RAM is not accessible) 1: EEPROM loaded correctly, the PDI operational (process data RAM is accessible)	R
1	PDIWDST	PDI Watchdog Timer Status 0: Timeout of the watchdog timer 1: Watchdog timer reloaded	R
2	ENHLINKD	Enhanced Link Detection Indication This bit is set to the value of bit 9 at address 0x0000 in the EEPROM the first time the EEPROM is loaded after power is initially supplied or after a reset. 0: Deactivated for all ports 1: Activated for at least one port	R
3	—	The read value is undefined.	R
4	PHYP0	Port 0 Link State Indication Physical link on port 0. 0: No link 1: Link detected	R
5	PHYP1	Port 1 Link State Indication Physical link on port 1. 0: No link 1: Link detected	R
6	PHYP2	Port 2 Link State Indication Port 2 is not available on this LSI. 0: No link 1: Link detected	R
7	PHYP3	Port 3 Link State Indication Port 3 is not available on this LSI. 0: No link 1: Link detected	R
8	LP0	Loop Port 0 State Indication 0: Open 1: Closed	R
9	COMP0	Port 0 Communication State Indication 0: No stable communication 1: Communication established	R

Bit	Symbol	Function	R/W
10	LP1	Loop Port 1 State Indication 0: Open 1: Closed	R
11	COMP1	Port 1 Communication State Indication 0: No stable communication 1: Communication established	R
12	LP2	Loop Port 2 State Indication Port 2 is not available on this LSI. 0: Open 1: Closed	R
13	COMP2	Port 2 Communication State Indication Port 2 is not available on this LSI. 0: No stable communication 1: Communication established	R
14	LP3	Loop Port 3 State Indication Port 3 is not available on this LSI. 0: Open 1: Closed	R
15	COMP3	Port 3 Communication State Indication Port 3 is not available on this LSI. 0: No stable communication 1: Communication established	R

Note: Reading this DL status register from the ECAT clears bit 2 of the ECAT event request register (ECAT_EVENT_REQ at 0x0210).
Avoid reading DL Status register from PDI.

The ESC_DL_STATUS register indicates the state of the EtherCAT slave controller.

36.8 Application Layer Registers

36.8.1 AL_CONTROL : AL Control Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0120

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	DEVIC EID	ERRIN DACK	INISTATE[3:0]			
Value after reset:	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	INISTATE[3:0]	Change the state transition of the device state machine. The access type from the EtherCAT master is R/W. 0x1: Initial state request 0x2: Pre-operational state request 0x3: Bootstrap state request 0x4: Safe-operational state request 0x8: Operational state request Others: Reserved	R
4	ERRINDACK	Error Indication Acknowledge (Response) The access type from the EtherCAT master is R/W. 0: Error Indication in AL status register is not acknowledged 1: Error Indication in AL status register is acknowledged	R
5	DEVICEID	Device ID Request The access type from the EtherCAT master is R/W. 0: No request is present 1: A request is present	R

Bit	Symbol	Function	R/W
15:6	—	The read values are undefined. The access type from the EtherCAT master is R/W. In this case, the write value should be 0.	R

Note: The PDI has to read the AL control register after the ECAT has written it. Otherwise, the ECAT cannot write again to the AL control register. Reading the AL control register from the PDI clears bit 0 of the AL event request register (AL_EVENT_REQ at 0x0220).

The AL_CONTROL register is used to change the state transition of the device state machine and to acknowledge error indication.

36.8.2 AL_STATUS : AL Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0130

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	DEVIC EID	ERR	ACTSTATE[3:0]			
Value after reset:	x	x	x	x	x	x	x	x	x	x	0	0	0	0	0	1

Bit	Symbol	Function	R/W
3:0	ACTSTATE[3:0]	State Machine State Indication Actual state of the device state machine The access type from the EtherCAT master is R. 0x1: Initial state 0x2: Pre-operational state 0x3: Request bootstrap state 0x4: Safe-operational state 0x8: Operational state	R/W
4	ERR	Error State Indication The access type from the EtherCAT master is R. 0: The device is in the state as requested or flag was cleared by command 1: The device has not entered the requested state or the state was changed as a result of local action	R/W
5	DEVICEID	Device ID Load State Indication The access type from the EtherCAT master is R. 0: Loading failed 1: Loading was successful	R/W
15:6	—	The read values are undefined. The write value should be 0.	R/W

Note: Reading this AL status register from the ECAT clears bit 3 of the ECAT event request register (ECAT_EVENT_REQ at 0x0210).
Avoid reading DL Status register from PDI.

The AL_STATUS register indicates the state of slave application.

36.8.3 AL_STATUS_CODE : AL Status Code Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0134

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	STATUSCODE[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	STATUSCODE[15:0]	AL status code The access type from the EtherCAT master is R.	R/W

The AL_STATUS_CODE register indicates an error code from slave application.

36.8.4 RUN_LED_OVERRIDE : RUN LED Override Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0138

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	OVER RIDEEN	LEDCODE[3:0]			

Value after reset: x x x 0 0 0 0 0

Bit	Symbol	Function	R/W
3:0	LEDCODE[3:0]	LED Code Indication (FSM state: Bits [3:0] of the AL Status register, AL_STATUS) 0x0: Off (FSM: 1-Init) 0xD: Blinking (FSM: 2-PreOp) 0xE: Flickering (FSM: 3-Bootstrap) 0xF: On (FSM: 8-Op) Others: Flash 1x - 12x (FSM: 4-SafeOp 1x)	R/W
4	OVERRIDEEN	Override Setting 0: Override disabled 1: Override enabled	R/W
7:5	—	The read values are undefined. The write value should be 0.	R/W

Note: Changing the value of the AL status register to an appropriate value will clear bit 4 (override enable).
Normally RUN LED is controlled by the AL status register (AL_STATUS at 0x0130) automatically. It is not necessary to override RUN LED in order to indicate the state of a general state machine.
For example, this register can be used to run special lighting patterns that indicate the positions of specific slaves.

The RUN_LED_OVERRIDE register is used to override control of the RUN LED pin.

36.8.5 ERR_LED_OVERRIDE : ERR LED Override Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0139

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	OVER RIDEEN	LEDCODE[3:0]			

Value after reset: x x x 0 0 0 0 0

Bit	Symbol	Function	R/W
3:0	LEDCODE[3:0]	LED Code Indication 0x0: Off 0xD: Blinking 0xE: Flickering 0xF: On Others: Flash 1x to 12x	R/W
4	OVERRIDEEN	Override Setting 0: Override disabled 1: Override enabled	R/W
7:5	—	The read values are undefined. The write value should be 0.	R/W

Note: Bit 4 (override enable) will be cleared if a new error occurs.
The ESC automatically controls an error LED under the conditions below. Regarding other errors, the error LED should be controlled by application using this register.

- SII EEPROM load error
- PDI watchdog timeout

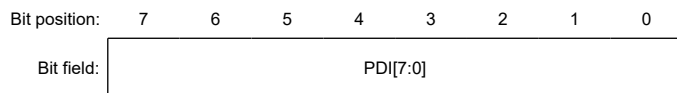
The ERR_LED_OVERRIDE register is used to override control of the error LED pin.

36.9 PDI Registers

36.9.1 PDI_CONTROL : PDI Control Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0140



Value after reset: 1 0 0 0 0 0 0 0

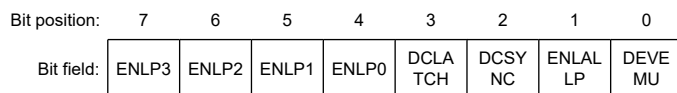
Bit	Symbol	Function	R/W
7:0	PDI[7:0]	Process Data Interface. In this LSI, the following value is indicated. 0x80: On-chip bus	R

This register indicates the type of PDI.

36.9.2 ESC_CONFIG : ESC Configuration Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0141



Value after reset: 0 0 0 0 1 1 0 0

Bit	Symbol	Function	R/W
0	DEVEMU	Device emulation (control of AL status) 0: The AL status register must be set by the PDI 1: The AL status register is set to a value written to the AL control register	R
1	ENLALLP	Sets enhanced link detection for all ports 0: Disabled (if bits 15 to 12 of address 0 in the EEPROM = 0) 1: Enabled at all ports	R
2	DCSYNC	Sets the SYNC output unit for distributed clocks (fixed to 1 in this LSI) 0: Disabled (power saving) 1: Enabled	R
3	DCLATCH	Sets the latch input unit for distributed clocks Fixed to 1 in this LSI. 0: Disabled (power saving) 1: Enabled	R
4	ENLP0	Port 0 Enhanced Link Detection Setting 0: Disabled (if bit 9 of address 0 in the EEPROM = 0) 1: Enabled	R
5	ENLP1	Port 1 Enhanced Link Detection Setting 0: Disabled (if bit 9 of address 0 in the EEPROM = 0) 1: Enabled	R
6	ENLP2	Port 2 Enhanced Link Detection Setting Port 2 is not available in this LSI. 0: Disabled (if bit 9 of address 0 in the EEPROM = 0) 1: Enabled	R

Bit	Symbol	Function	R/W
7	ENLP3	Port 3 Enhanced Link Detection Setting Port 3 is not available in this LSI. 0: Disabled (if bit 9 of address 0 in the EEPROM = 0) 1: Enabled	R

The ESC_CONFIG register indicates configuration of the EtherCAT slave controller.

36.9.3 PDI_CONFIG : PDI Configuration Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0150

Bit position: 7 6 5 4 3 2 1 0

Bit field:	ONCHIPBUS[2:0]	ONCHIPBUSCLK[4:0]
------------	----------------	-------------------

Value after reset: 0 1 0 0 0 1 0 0

Bit	Symbol	Function	R/W
4:0	ONCHIPBUSCLK [4:0]	On-Chip Bus Clock Indication Indicate the frequency of the on-chip bus clock. In this LSI, the value is always 0x04 (corresponding to 100 MHz).	R
7:5	ONCHIPBUS[2:0]	On-Chip Bus Type Indication Indicate the type of on-chip bus. In this LSI, the value is always 010b.	R

The PDI_CONFIG register indicates configuration of the PDI.

36.9.4 SYNC_LATCH_CONFIG : SYNC/LATCH PDI Configuration Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0151

Bit position: 7 6 5 4 3 2 1 0

Bit field:	SYNC 1MAP	SYNC LAT1	SYNC1OUT [1:0]	SYNC 0MAP	SYNC LAT0	SYNC0OUT [1:0]
------------	--------------	--------------	-------------------	--------------	--------------	-------------------

Value after reset: 1 1 1 0 1 1 1 0

Bit	Symbol	Function	R/W
1:0	SYNC0OUT[1:0]	SYNC0 Output Driver and Polarity Indication Indicate the SYNC0 output driver/polarity. In this LSI, the value is always 10b (push-pull and active-high).	R
2	SYNCLAT0	SYNC0/LATCH0 Indication Indicates the SYNC0/LATCH0 configuration. In this LSI, the value is always 1.*1 0: LATCH0 input 1: SYNC0 output	R
3	SYNC0MAP	SYNC0 State Mapping Indication Indicates enabling or disabling of mapping of the SYNC0 state to bit 2 of the AL event request register (AL_EVENT_REQ at 0x0220). This is always enabled in this LSI, so the value indicated is always 1 (enabled). 0: Disabled 1: Enabled	R
5:4	SYNC1OUT[1:0]	SYNC1 Output Driver and Polarity Indication Indicate the SYNC1 output driver/polarity. In this LSI, the value is always 10b (push-pull and active-high).	R

Bit	Symbol	Function	R/W
6	SYNCLAT1	SYNC1/LATCH1 Indication Indicates the SYNC1/LATCH1 configuration. In this LSI, the value is always 1.*1 0: LATCH1 input 1: SYNC1 output	R
7	SYNC1MAP	SYNC1 State Mapping Indication Indicates enabling or disabling of mapping of the SYNC1 state to bit 3 of the AL event request register (AL_EVENT_REQ at 0x0220). This is always enabled in this LSI, so the value indicated is always 1 (enabled). 0: Disabled 1: Enabled	R

Note 1. Latch input is available though the value indicates SYNC output. Use the MPC function in order to switch SYNC output to LATCH input and vice versa.

The SYNC_LATCH_CONFIG register indicates the configuration of SYNC output and LATCH input.

36.9.5 EXT_PDI_CONFIG : Extended PDI Configuration Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0152

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	DATABUSWID [1:0]
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0	0

Bit	Symbol	Function	R/W
1:0	DATABUSWID[1:0]	PDI Data Bus Width Indication Indicate the data bus width of the PDI. In this LSI, the value is always 00b (4 bytes). 0 0: 4 bytes 0 1: 1 byte 1 0: 2 bytes 1 1: Reserved	R
15:2	—	The read values are undefined.	R

The EXT_PDI_CONFIG register indicates configuration of the PDI.

36.10 Interrupt Registers

36.10.1 ECAT_EVENT_MASK : ECAT Event Mask Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0200

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	ECATEVMASK[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	ECATEVMASK[15:0]	Event Request Mask Setting The access type from the EtherCAT master is R/W. 0: The corresponding bit of the ECAT Event Request register (ECAT_EVENT_REQ at 0x0210) is not mapped 1: The corresponding bit of the ECAT Event Request register is mapped	R

The ECAT Event Request (ECAT interrupt) is used to transmit the slave event to the EtherCAT master. This register is used to set mask to each event of the ECAT Event Request register (ECAT_EVENT_REQ at 0x0210). The logical AND of each effective bit in the ECAT Event Request register and the corresponding bit of this register is taken and the result produces the interrupt signal.

36.10.2 AL_EVENT_MASK : AL Event Mask Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0204

Bit position: 31 0

Bit field: ALEVMASK[31:0]

Value after reset: 0 0 0 0 0 0 0 0 0 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 1 0 0 0 0 1 1 1 1

Bit	Symbol	Function	R/W
31:0	ALEVMASK[31:0]	Event Request Mask Setting The access type from the EtherCAT master is R. 0: The corresponding bit of the AL Event Request register (AL_EVENT_REQ at 0x0220) is not mapped 1: The corresponding bit of the AL Event Request register is mapped	R/W

The AL event request (PDI interrupt) is used to transmit the ESC interrupt to the slave application. This register is used to set mask to each event of the AL Event Request register (AL_EVENT_REQ at 0x0220). The logical AND of each effective bit in the AL Event Request register and the corresponding bit of this register is taken and the result produces the interrupt signal.

36.10.3 ECAT_EVENT_REQ : ECAT Event Request Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0210

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: — — — — SMST A7 SMST A6 SMST A5 SMST A4 SMST A3 SMST A2 SMST A1 SMST A0 ALSTA DLSTA — DCLATCH

Value after reset: x x x x 0 0 0 0 0 0 0 0 0 0 0 0 x 0

Bit	Symbol	Function	R/W
0	DCLATCH	DC Latch Event State Indication This bit is cleared by reading DC latch event times from the ECAT for ECAT controlled latch units, so the latch 0 and 1 status registers (DC_LATCH_STAT0 and DC_LATCH_STAT1 at 0x09AE and 0x09AF) indicate no event. 0: No change on DC latch inputs 1: At least one change on DC latch inputs	R
1	—	The read value is undefined.	R
2	DLSTA	DL Status Event State Indication This bit is cleared by reading out the DL status register (ESC_DL_STATUS at 0x0110 or 0x0111) from the ECAT. 0: No change in DL status 1: DL status change	R
3	ALSTA	AL Status Event State Indication This bit is cleared by reading out the AL status register (AL_STATUS at 0x0130 or 0x0131) from the ECAT. 0: No change in AL status 1: AL status change	R

Bit	Symbol	Function	R/W
4	SMSTA0	Mirror value of SyncManager 0 Status Indication 0: No Sync channel 0 event 1: Sync channel 0 event pending	R
5	SMSTA1	Mirror value of SyncManager 1 Status Indication 0: No Sync channel 1 event 1: Sync channel 1 event pending	R
6	SMSTA2	Mirror value of SyncManager 2 Status Indication 0: No Sync channel 2 event 1: Sync channel 2 event pending	R
7	SMSTA3	Mirror value of SyncManager 3 Status Indication 0: No Sync channel 3 event 1: Sync channel 3 event pending	R
8	SMSTA4	Mirror value of SyncManager 4 Status Indication 0: No Sync channel 4 event 1: Sync channel 4 event pending	R
9	SMSTA5	Mirror value of SyncManager 5 Status Indication 0: No Sync channel 5 event 1: Sync channel 5 event pending	R
10	SMSTA6	Mirror value of SyncManager 6 Status Indication 0: No Sync channel 6 event 1: Sync channel 6 event pending	R
11	SMSTA7	Mirror value of SyncManager 7 Status Indication 0: No Sync channel 7 event 1: Sync channel 7 event pending	R
15:12	—	The read values are undefined.	R

The ECAT_EVENT_REQ register indicates the source of ECAT event requests (ECAT interrupts).

36.10.4 AL_EVENT_REQ : AL Event Request Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0220

Bit position:	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x
Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SMINT 7	SMINT 6	SMINT 5	SMINT 4	SMINT 3	SMINT 2	SMINT 1	SMINT 0	—	WDPD	—	SYNC ACT	DCSY NC1S TA	DCSY NC0S TA	DCLA TCH	ALCT RL
Value after reset:	0	0	0	0	0	0	0	0	x	0	x	0	0	0	0	0

Bit	Symbol	Function	R/W
0	ALCTRL	AL Control Event State Indication This bit is cleared by reading the AL control register (AL_CONTROL at 0x0120 or 0x0121) from the PDI. 0: No change in the AL control register 1: The AL control register has been written	R
1	DCLATCH	DC Latch Event State Indication This bit is cleared by reading DC latch event times from the PDI for PDI controlled latch units, so the latch 0 and 1 status registers (DC_LATCH_STAT0 and DC_LATCH_STAT1 at 0x09AE and 0x09AF) indicate no event. 0: No change on DC latch inputs 1: At least one change on DC latch inputs	R

Bit	Symbol	Function	R/W
2	DCSYNC0STA	DC SYNC0 State Indication This bit is cleared by reading the SYNC0 status register (DC_SYNC0_STAT at 0x098E) from the PDI.	R
3	DCSYNC1STA	DC SYNC1 State Indication This bit is cleared by reading the SYNC1 status register (DC_SYNC1_STAT at 0x098F) from the PDI.	R
4	SYNCACT	SyncManager Activation Indication Change of the SyncManager activation register (SMn.ACT at 0x0806 + 0x8 × n) This bit is cleared by reading SyncManager activation registers from the PDI. 0: No change in any SyncManager 1: At least one SyncManager changed	R
5	—	The read value is undefined.	R
6	WDPD	Watchdog Process Data Indication This bit is cleared by reading the watchdog status process data register (WDS_DATA at 0x0440) from the PDI. 0: Valid 1: Timeout	R
7	—	The read value is undefined.	R
8	SMINT0	SyncManager 0 interrupt (bit 0 or 1 of the SyncManager status register (0x0805)) 0: No SyncManager 0 interrupt 1: SyncManager 0 interrupt pending	R
9	SMINT1	SyncManager 1 interrupt (bit 0 or 1 of the SyncManager status register (0x080D)) 0: No SyncManager 1 interrupt 1: SyncManager 1 interrupt pending	R
10	SMINT2	SyncManager 2 interrupt (bit 0 or 1 of the SyncManager status register (0x0815)) 0: No SyncManager 2 interrupt 1: SyncManager 2 interrupt pending	R
11	SMINT3	SyncManager 3 interrupt (bit 0 or 1 of the SyncManager status register (0x081D)) 0: No SyncManager 3 interrupt 1: SyncManager 3 interrupt pending	R
12	SMINT4	SyncManager 4 interrupt (bit 0 or 1 of the SyncManager status register (0x0825)) 0: No SyncManager 4 interrupt 1: SyncManager 4 interrupt pending	R
13	SMINT5	SyncManager 5 interrupt (bit 0 or 1 of the SyncManager status register (0x082D)) 0: No SyncManager 5 interrupt 1: SyncManager 5 interrupt pending	R
14	SMINT6	SyncManager 6 interrupt (bit 0 or 1 of the SyncManager status register (0x0835)) 0: No SyncManager 6 interrupt 1: SyncManager 6 interrupt pending	R
15	SMINT7	SyncManager 7 interrupt (bit 0 or 1 of the SyncManager status register (0x083D)) 0: No SyncManager 7 interrupt 1: SyncManager 7 interrupt pending	R
31:16	—	The read values are undefined.	R

The AL_EVENT_REQ register indicates the source of AL event requests (PDI interrupts).

36.11 Error Counter Registers

36.11.1 RX_ERR_COUNTn : RX Error Counter n Register (n = 0 to 1)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0300 + 0x2 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	RXERRCNT[7:0]								INVFRMCNT[7:0]							
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	INVFRMCNT[7:0]	Invalid Frame Counter Value Indication Counter value of invalid frames for port n counting is stopped when 0xFF is reached. These bits are cleared if one of the RX error counters (RX_ERR_COUNTn, FWD_RX_ERR_COUNTn) is written. The access type from the EtherCAT master is R/W.	R
15:8	RXERRCNT[7:0]	RX Frame Error Counter Value Indication Counter value of RX errors for port n counting is stopped when 0xFF is reached. The number of RX errors of MII interface is counted. These bits are cleared if one of the RX error counters (RX_ERR_COUNTn, FWD_RX_ERR_COUNTn) is written. The access type from the EtherCAT master is R/W.	R

The RX_ERR_COUNTn register counts RX frame errors.

36.11.2 FWD_RX_ERR_COUNTn : Forwarded RX Error Counter n Register (n = 0 to 1)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0308 + 0x1 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	FWDERRCNT[7:0]							
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	FWDERRCNT[7:0]	Forwarded Error Counter Value Indication Counter value of forwarded RX error frames for port n counting is stopped when 0xFF is reached. These bits are cleared if one of the RX error counters (RX_ERR_COUNTn, FWD_RX_ERR_COUNTn) is written. The access type from the EtherCAT master is R/W.	R

The FWD_RX_ERR_COUNTn register counts forwarded RX frame errors.

36.11.3 ECAT_PROC_ERR_COUNT : ECAT Processing Unit Error Counter Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x030C

Bit position:	7	6	5	4	3	2	1	0
Bit field:	EPUERRCNT[7:0]							
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	EPUEERRCNT[7:0]	Processing Unit Error Counter Value Indication ECAT processing unit error counter value counting is stopped when 0xFF is reached. This register counts errors of frames passing the processing unit. Writing to this register clears it. The access type from the EtherCAT master is R/W.	R

The ECAT_PROC_ERR_COUNT register counts frame errors passing the ECAT processing unit.

36.11.4 PDI_ERR_COUNT : PDI Error Counter Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x030D

Bit position: 7 6 5 4 3 2 1 0

Bit field: PDIERRCNT[7:0]

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	PDIERRCNT[7:0]	PDI Error Counter Value Indication Counting is stopped when 0xFF is reached. Counting starts when an interface error occurs due to access to the PDI. Writing to this register clears it. The access type from the EtherCAT master is R/W.	R

The PDI_ERR_COUNT register counts PDI access errors.

36.11.5 LOST_LINK_COUNTn : Lost Link Counter n Register (n = 0 to 1)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0310 + 0x1 × n

Bit position: 7 6 5 4 3 2 1 0

Bit field: LOSTLINKCNT[7:0]

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	LOSTLINKCNT[7:0]	Lost Link Counter Value Indication Lost link counter value for port n counting is stopped when 0xFF is reached. Counting starts only when port loop is Auto or Auto-Close. Only lost links at open ports are counted. Writing to one of the lost link counter registers clears it. The access type from the EtherCAT master is R/W.	R

The LOST_LINK_COUNTn register counts lost links at port n.

36.12 Watchdog Registers

36.12.1 WD_DIVIDE : Watchdog Divider Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0400

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WDDIV[15:0]															
Value after reset:	0	0	0	0	1	0	0	1	1	1	0	0	0	0	1	0

Bit	Symbol	Function	R/W
15:0	WDDIV[15:0]	Watchdog Clock Frequency Divisor Setting Set the frequency divisor of the clock to drive counting by the watchdog timer in units of ticks at 25 MHz. The clock that drives counting by the watchdog timer is obtained by dividing 25 MHz by the value in this register plus 2. The default value is 2498, which corresponds to a period of 100 μs. The access type from the EtherCAT master is R/W.	R

The WD_DIVIDE register is used to set the ratio for dividing 25 MHz to obtain the basic period for incrementing the watchdog timer.

36.12.2 WDT_PDI : Watchdog Time PDI Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0410

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WDTIMPDI[15:0]															
Value after reset:	0	0	0	0	0	0	1	1	1	1	1	0	1	0	0	0

Bit	Symbol	Function	R/W
15:0	WDTIMPDI[15:0]	Watchdog Overflow Time Setting Set the time until the PDI watchdog timer overflows as a number of times the watchdog is incremented. With the default values for these bits and the setting of the watchdog divider, the time for a single incrementation is 100 μs, so the watchdog timer overflows when $100 \mu\text{s} \times 1000 = 100 \text{ ms}$ elapsed. Setting these bits to 0 disables the watchdog timer. Access to the PDI restarts the watchdog timer. The access type from the EtherCAT master is R/W.	R

The WDT_PDI register is used to set the time until the PDI watchdog timer overflows.

36.12.3 WDT_DATA : Watchdog Time Process Data Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0420

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	WDTIMPD[15:0]															
Value after reset:	0	0	0	0	0	0	1	1	1	1	1	0	1	0	0	0

Bit	Symbol	Function	R/W
15:0	WDTIMPD[15:0]	Watchdog Overflow Time Setting Set the time until the process data watchdog timer overflows as a number of times the watchdog is incremented. With the default values for these bits and the setting of the watchdog divider, the time for a single incrementation is 100 μs, so the watchdog timer overflows when $100 \mu\text{s} \times 1000 = 100 \text{ ms}$ elapsed. There is one watchdog for all SyncManagers. Setting these bits to 0 disables the watchdog timer. Access to the Watchdog Trigger Enable bit of SyncManager restarts the watchdog timer. The access type from the EtherCAT master is R/W.	R

The WDT_DATA register is used to set the time until the process data watchdog timer overflows.

36.12.4 WDS_DATA : Watchdog Status Process Data Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0440

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	—	—	—	—	—	—	—	—	WDST APD
Value after reset:	x	x	x	x	x	x	x	x	x	x	x	x	x	x	x	0

Bit	Symbol	Function	R/W
0	WDSTAPD	Watchdog State Indication Indicates the state of the process data watchdog timer triggered by SyncManagers. Reading this register clears bit 6 of the AL Event Request register (AL_EVENT_REQ at 0x0220). 0: The timeout period of the process data watchdog timer elapses 1: The process data watchdog timer is active or disabled	R
15:1	—	The read values are undefined.	R

The WDS_DATA register indicates the state of the process data watchdog timer.

36.12.5 WDC_DATA : Watchdog Counter Process Data Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0442

Bit position:	7	6	5	4	3	2	1	0
Bit field:	WDCNTPD[7:0]							
Value after reset:	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
7:0	WDCNTPD[7:0]	Watchdog Counter Value Indication Counter value of the process data watchdog timer counting stops when 0xFF is reached. Counting starts on a timeout of the process data watchdog timer. Writing to one of the watchdog counter registers (WDC_DATA, WDC_PDI at 0x0442 and 0x0443) clears the counter. The access type from the EtherCAT master is R/W.	R

The WDC_DATA register indicates the timeout counter value of the process data watchdog timer.

36.12.6 WDC_PDI : Watchdog Counter PDI Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0443

Bit position: 7 6 5 4 3 2 1 0

Bit field: WDCNTDPDI[7:0]

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
7:0	WDCNTDPDI[7:0]	Watchdog Counter Value Indication Counter value of the PDI watchdog timer counting stops when 0xFF is reached. Counting starts on a timeout of the PDI watchdog timer. Writing to one of the watchdog counter registers (WDC_DATA, WDC_PDI at 0x0442 and 0x0443) clears the counter. The access type from the EtherCAT master is R/W.	R

The WDC_PDI register indicates the timeout counter value of the PDI watchdog timer.

36.13 SII EEPROM Interface Registers

The EtherCAT controls the SII EEPROM interface if bit 0 is 0 in the EEPROM Configuration register (EEP_CONF at 0x0500) and in the EEPROM PDI Access State register (EEP_STATE at 0x0501). Otherwise, the PDI controls the EEPROM interface.

36.13.1 EEP_CONF : EEPROM Configuration Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0500

Bit position: 7 6 5 4 3 2 1 0

Bit field: — — — — — — FORC EECAT CTRL PDI

Value after reset: x x x x x x 0 0

Bit	Symbol	Function	R/W
0	CTRLPDI	PDI EEPROM Control Specifies whether EEPROM control is offered to the PDI. The access type from the EtherCAT master is R/W. 0: The PDI has no EEPROM control 1: The PDI has EEPROM control	R
1	FORCEECAT	EEPROM Access Right Change Forcibly changes the right of access to the EEPROM by the ECAT. The access type from the EtherCAT master is R/W. 0: No change 1: Reset bit 0 of the EEPROM PDI Access State register (EEP_STATE at 0x0501) to 0. That is, prohibit access to the EEPROM by the PDI.	R
7:2	—	The read values are undefined.	R

The EEP_CONF register is used to configure EEPROM access.

36.13.2 EEP_STATE : EEPROM PDI Access State Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0501

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	PDIACCESS
Value after reset:	x	x	x	x	x	x	x	0

Bit	Symbol	Function	R/W
0	PDIACCESS	EEPROM Access Right Setting Write access from the PDI is only possible when bit 0 is 1 and bit 1 is 0 in the EEPROM Configuration register (EEP_CONF at 0x0500). The access type from the EtherCAT master is R. 0: Prohibits the PDI from access to the EEPROM 1: The PDI has access to the EEPROM	R/W
7:1	—	The read values are undefined. The write value should be 0. The access type from the EtherCAT master is R.	R/W

The EEP_STATE register is used to set the right of access to the EEPROM by the PDI.

36.13.3 EEP_CONT_STAT : EEPROM Control/Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0502

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	BUSY	WREN ERR	ACKC MDER R	LOAD STA	CKSU MERR	COMMAND[2:0]			PROM SIZE	READ BYTE	—	—	—	—	—	ECAT WREN
Value after reset:	0	0	0	0	0	0	0	0	0/1 ^{*1}	0	x	x	x	x	x	0

Bit	Symbol	Function	R/W
0	ECATWREN	ECAT Write Enable ^{*3} This bit is always 1 if the PDI has EEPROM control. The access type from the EtherCAT master is R/W. 0: Write requests are disabled 1: Write requests are enabled	R
5:1	—	The read values are undefined. The write value should be 0.	R/W
6	READBYTE	EEPROM Read Byte Indication Indicates supported EEPROM read bytes. 0: 4 bytes 1: 8 bytes	R
7	PROMSIZE	EEPROM Algorithm Indication Indicates the selected EEPROM algorithm. 0: 1 address byte (1-Kbit to 16-Kbit EEPROMs) 1: 2 address bytes (32-Kbit to 4-Mbit EEPROMs)	R
10:8	COMMAND[2:0]	Command ^{*3} Write: Initiates the commands below. Read: Indicates the currently executed command. Commands: 0 0 0: No command/EEPROM idle (clear error bits) 0 0 1: Read 0 1 0: Write 1 0 0: Reload Others: Reserved/invalid commands (must not be issued)	R/W

Bit	Symbol	Function	R/W
11	CKSUMERR	Checksum Error Indication Indicates checksum error in the ESC configuration area. 0: No error in the checksum 1: Error in the checksum	R
12	LOADSTA	EEPROM Loading Status Indication Indicates EEPROM loading status. 0: EEPROM has been loaded and device information has no problem 1: EEPROM has not been loaded and device information is not available (EEPROM loading in progress or finished with a failure)	R
13	ACKCMDERR	Acknowledge/Command Error Indication Indicates acknowledge/command error.*2 0: No error 1: Missing EEPROM acknowledge or invalid command	R
14	WRENERR	Write Enable Error Indication Indicates write enable error.*2 0: No error 1: Write command without write enable	R
15	BUSY	EEPROM Interface State Indication Indicates a busy state of the EEPROM interface. 0: The EEPROM interface is idle 1: The EEPROM interface is busy	R

Note: Write access depends on the assignment of the EEPROM interface (ECAT/PDI). Write access is generally blocked if the EEPROM interface is busy (bit 15 = 1).

Note 1. This initial value will be changed by setting of EEPROMSIZE bit in EtherCAT Operation Mode Register (ECATOPMOD).

Note 2. Error bits are cleared by writing 000b (or any valid command) to command bits [10:8].

Note 3. ECAT write enable bit 0 is self-cleared at the SOF of the next frame. Command bits [10:8] are also self-cleared after the command is executed (EEPROM busy ends).

Writing 000b to command bits [10:8] also clears the error bits 14 and 13. Command bits [10:8] are ignored if the acknowledge/command error bit 13 is 1.

The EEP_CONT_STAT register is used to control access to the EEPROM and indicate the status.

36.13.4 EEP_ADR : EEPROM Address Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0504

Bit position: 31

0

Bit field:

ADDRESS[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	ADDRESS[31:0]	EEPROM Address Setting Actually used EEPROM address bits: [9:0]: EEPROM size of up to 16 Kbits [17:0]: EEPROM size of 32 Kbits to 4 Mbits 0: First word (= 16 bits) 1: Second word ⋮	R/W

Note: Write access depends on the assignment of the EEPROM interface (ECAT/PDI). Write access is generally blocked if the EEPROM interface is busy (bit 15 is 1 in the EEPROM Control/Status register, EEP_CONT_STAT, at 0x0502).

The EEP_ADR register is used to set the EEPROM address to be accessed.

36.13.5 EEP_DATA : EEPROM Data Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0508

Bit position: 31 16 15 0

Bit field:

HIDATA[15:0]	LODATA[15:0]
--------------	--------------

Value after reset: 0

Bit	Symbol	Function	R/W
15:0	LODATA[15:0]	Data to be written to the EEPROM or data read from the EEPROM (lower 2 bytes)	R/W
31:16	HIDATA[15:0]	Data read from the EEPROM (upper 2 bytes)	R

Note: Write access depends on the assignment of the EEPROM interface (ECAT/PDI). Write access is generally blocked if the EEPROM interface is busy (bit 15 is 1 in the EEPROM Control/Status register, EEP_CONT_STAT, at 0x0502).

The EEP_DATA register is used to set write data to the EEPROM or indicates read data from the EEPROM. It can be written in 1-word units and read in 2-word units.

36.14 MII Management Interface Registers

36.14.1 MII_CONT_STAT : MII Management Control/Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0510

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field:

BUSY	CMDE RR	READ ERR	—	—	—	COMMAND[1:0]	PHYOFFSET[4:0]	MILIN K	PDICT RL	WREN
------	------------	-------------	---	---	---	--------------	----------------	------------	-------------	------

Value after reset: 0 0 0 x x x 0 0 0 0 0 0 0 0 1 0

Bit	Symbol	Function	R/W
0	WREN	Write Enable This bit is always 1 if the PDI controls the MII management interface. The access type from the EtherCAT master is R/W. 0: Disabled 1: Enabled	R
1	PDICTRL	PDI Control Indication Indicates whether the MII management interface can be controlled by the PDI. The interface is controlled by the MII management ECAT Access State register (MII_ECAC_ACS_STAT at 0x0516) and the MII management PDI Access State register (MII_PDI_ACS_STAT at 0x0517). 0: Only ECAT control 1: PDI control possible	R
2	MILINK	MI Link Detection 0: Not available 1: Available	R
7:3	PHYOFFSET[4:0]	PHY Address Offset Indication Indicate the PHY address offset.	R

Bit	Symbol	Function	R/W
9:8	COMMAND[1:0]	Command Write: Initiates the commands below. Read: Indicates the currently executed command. Commands: 0 0: No command/MII idle (clear error bits) 0 1: Read 1 0: Write Others: Setting prohibited	R/W
12:10	—	The read values are undefined. The write value should be 0.	R/W
13	READERR	Read Error Indication Indicates whether a read error occurred. This bit is cleared by writing to this register. 0: No read error 1: Read error occurred (PHY or register not available)	R/W
14	CMDERR	Command Error Indication Indicates whether a command error occurred. This bit is cleared by executing a valid command or writing 00b to COMMAND[1:0]. 0: Last command was successful 1: Invalid command or write command without write enable	R
15	BUSY	MII Management State Indication Indicates that the MII management interface is busy. 0: MII management interface is idle 1: MII management interface is busy	R

The MII_CONT_STAT register is used to control the MII management interface and to indicate the status.

Write enable bit 0 is self-cleared at the SOF of the next frame. Command bits [9:8] are also self-cleared after the command is executed (busy ends). Writing 00b to command bits also clears the error bits 14 and 13. Command bits are cleared after the command is executed.

Write access depends on the assignment of the management interface (ECAT/PDI). Write access is generally blocked if the management interface is busy (bit 15 = 1 in this register).

36.14.2 PHY_ADR : PHY Address Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0512

Bit position: 7 6 5 4 3 2 1 0

Bit field:	—	—	—	PHYADDR[4:0]			
------------	---	---	---	--------------	--	--	--

Value after reset: x x x 0 0 0 0 0

Bit	Symbol	Function	R/W
4:0	PHYADDR[4:0]	PHY Address Setting	R/W
7:5	—	The read values are undefined.	R

The PHY_ADR register is used to set the PHY address.

Write access depends on the assignment of the management interface (ECAT/PDI). Write access is generally blocked if the management interface is busy (bit 15 is 1 in the MII management control/status register, MII_CONT_STAT, at 0x0510).

36.14.3 PHY_REG_ADR : PHY Register Address Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0513

Bit position: 7 6 5 4 3 2 1 0

Bit field: — — — PHYREGADDR[4:0]

Value after reset: x x x 0 0 0 0 0

Bit	Symbol	Function	R/W
4:0	PHYREGADDR[4:0]	Address of PHY register	R/W
7:5	—	The read values are undefined.	R/W

The PHY_REG_ADR register is used to set the PHY register address.

Write access depends on the assignment of the management interface (ECAT/PDI). Write access is generally blocked if the management interface is busy (bit 15 is 1 in the MII management Control/Status register, MII_CONT_STAT, at 0x0510).

36.14.4 PHY_DATA : PHY Data Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0514

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: PHYREGDATA[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	PHYREGDATA[15:0]	PHY Register Data Indication/Setting	R/W

The PHY_DATA register is used to set data to write to PHY registers or to indicate data read from PHY registers.

Write access depends on the assignment of the management interface (ECAT/PDI). Write access is generally blocked if the management interface is busy (bit 15 is 1 in the MII management Control/Status register, MII_CONT_STAT, at 0x0510).

36.14.5 MII_ECAT_ACS_STAT : MII Management ECAT Access State Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0516

Bit position: 7 6 5 4 3 2 1 0

Bit field: — — — — — — ACSMI

Value after reset: x x x x x x x 0

Bit	Symbol	Function	R/W
0	ACSMII	MII Management Interface Access Right Setting Right of access to the MII management interface The access type from the EtherCAT master is R/W. 0: Enables access to the MII management interface by the PDI. 1: Exclusive access to the MII management interface by the ECAT	R
7:1	—	The read values are undefined.	R

Note: Write access is only possible when bit 0 is 1 in the MII management PDI Access State register (MII_PDI_ACS_STAT at 0x0517).

The MII_ECAT_ACS_STAT register is used to set the right of access to the MII management interface.

36.14.6 MII_PDI_ACS_STAT : MII Management PDI Access State Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0517

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	FORP DI	ACSMI I
Value after reset:	x	x	x	x	x	x	0	0

Bit	Symbol	Function	R/W
0	ACSMII	Right of access to the MII management interface The access type from the EtherCAT master is R. 0: Access to the MII management interface by the ECAT 1: Access to the MII management interface by the PDI	R/W
1	FORPDI	Forced change of access by the PDI (forced change of bit 0) The access type from the EtherCAT master is R/W. 0: The value of bit 0 of this register is not changed 1: The value of bit 0 of this register is reset to 0 (the right of access is changed to the ECAT)	R
7:2	—	The read values are undefined. The access type from the EtherCAT master is R.	R/W

Note: Write access to bit 0 from the PDI is only possible if the following two conditions are satisfied.

- Bit 0 is 0 in the MII management ECAT Access State register (MII_ECAT_ACS_STAT at 0x0516).
- Bit 1 is 0 in the MII management PDI Access State register (MII_PDI_ACS_STAT at 0x0517).

The MII_PDI_ACS_STAT register is used to set the right of access to the MII management interface.

36.15 FMMU Registers

36.15.1 FMMUn_L_START_ADR : FMMU Logical Start Address n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0600 + 0x10 × n

Bit position:	31	0
Bit field:	LSTAADR[31:0]	
Value after reset:	0 0	

Bit	Symbol	Function	R/W
31:0	LSTAADR[31:0]	Logical Start Address Setting Set the start of the logical address within the EtherCAT address space. The access type from the EtherCAT master is R/W.	R

The FMMUn_L_START_ADR register is used to set the logical start address within the EtherCAT address space for FMMU.

36.15.2 FMMUn_LEN : FMMU Length n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0604 + 0x10 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	FMMULEN[15:0]															
Value after reset:	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	Symbol	Function	R/W
15:0	FMMULEN[15:0]	Area Size Specification Set the area size in byte units. Address at the end of the logical address range set by the FMMU minus the address at the start of the logical address range set by the FMMU plus 1. The access type from the EtherCAT master is R/W.	R

The FMMUn_LEN register is used to set the length for FMMU area in byte units.

36.15.3 FMMUn_L_START_BIT : FMMU Logical Start Bit n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0606 + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	LSTABIT[2:0]		
Value after reset:	x	x	x	x	x	0	0	0

Bit	Symbol	Function	R/W
2:0	LSTABIT[2:0]	Start Bit Setting Set the start bits of the logical start address for FMMU. The access type from the EtherCAT master is R/W.	R
7:3	—	The read values are undefined.	R

The FMMUn_L_START_BIT register is used to set the start bits of the logical start address for FMMU.

36.15.4 FMMUn_L_STOP_BIT : FMMU Logical Stop Bit n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0607 + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	LSTPBIT[2:0]		
Value after reset:	x	x	x	x	x	0	0	0

Bit	Symbol	Function	R/W
2:0	LSTPBIT[2:0]	Last Bit Setting Set the last bits of the logical end address for FMMU. The access type from the EtherCAT master is R/W.	R
7:3	—	The read values are undefined.	R

The FMMUn_L_STOP_BIT register is used to set the last bits of the logical end address for FMMU.

36.15.5 FMMUn_P_START_ADR : FMMU Physical Start Address n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0608 + 0x10 × n

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: PHYSTAADR[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	PHYSTAADR[15:0]	Physical Start Address Setting Set the physical start address to which the logical start address will be mapped. The address is set as an offset from the base address (0x8013_0000). The access type from the EtherCAT master is R/W.	R

The FMMUn_P_START_ADR register is used to set the physical start address of the ESC to which the logical start address will be mapped by the FMMU.

36.15.6 FMMUn_P_START_BIT : FMMU Physical Start Bit n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x060A + 0x10 × n

Bit position: 7 6 5 4 3 2 1 0

Bit field: — — — — — PHYSTABIT[2:0]

Value after reset: x x x x x 0 0 0

Bit	Symbol	Function	R/W
2:0	PHYSTABIT[2:0]	Physical Start Bit Setting Set the start bits of the physical start address to which the start bits of the logical start address will be mapped. The access type from the EtherCAT master is R/W.	R
7:3	—	The read values are undefined.	R

The FMMUn_P_START_BIT register is used to set the start bits of the physical start address of the ESC to which the start bits of the logical start address will be mapped by the FMMU.

36.15.7 FMMUn_TYPE : FMMU Type n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x060B + 0x10 × n

Bit position: 7 6 5 4 3 2 1 0

Bit field: — — — — — WRIT E READ

Value after reset: x x x x x x 0 0

Bit	Symbol	Function	R/W
0	READ	Read Access Mapping Setting The access type from the EtherCAT master is R/W. 0: Disabled 1: Enabled	R

Bit	Symbol	Function	R/W
1	WRITE	Write Access Mapping Setting The access type from the EtherCAT master is R/W. 0: Disabled 1: Enabled	R
7:2	—	The read values are undefined.	R

The FMMUn_TYPE register is used to set the type of FMMU access.

36.15.8 FMMUn_ACT : FMMU Activate n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x060C + 0x10 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	ACTIV ATE

Value after reset: x x x x x x x 0

Bit	Symbol	Function	R/W
0	ACTIVATE	FMMU Enable/Disable The access type from the EtherCAT master is R/W. 0: Disabled 1: Enabled	R
7:1	—	The read values are undefined.	R

The FMMUn_ACT register is used to enable or disable FMMU.

36.16 SyncManager Registers

36.16.1 SMn_P_START_ADR : SyncManager Physical Start Address n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0800 + 0x8 × n

Bit position:	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
Bit field:	SMSTAADDR[15:0]															

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	SMSTAADDR[15:0]	Physical Start Address Setting Specify the physical start address for the area assigned to SyncManager. The access type from the EtherCAT master is R/W.	R

Note: The SMn_P_START_ADR register can only be written when SyncManager is disabled (bit 0 is 0 in the SyncManager Activate n register (SMn.ACT at 0x0806 + 0x8 × n).

This register is used to set the physical start address for the area assigned to SyncManager.

36.16.2 SMn_LEN : SyncManager Length n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0802 + 0x8 × n

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: SMLen[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	SMLen[15:0]	Area Size Setting Set the number of bytes assigned to SyncManager. Set a value greater than 1. Otherwise, SyncManager is not activated. The access type from the EtherCAT master is R/W.	R

Note: This register can only be written when SyncManager is disabled (bit 0 is 0 in the SyncManager Activate n register (SMn.ACT at 0x0806 + 0x8 × n)).

The SMn_LEN register is used to set the size of the area assigned to SyncManager.

36.16.3 SMn_CONTROL : SyncManager Control n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0804 + 0x8 × n

Bit position: 7 6 5 4 3 2 1 0

Bit field: — WDTR GEN IRQPD I IRQEC AT DIR[1:0] OPEMODE[1:0]

Value after reset: x 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
1:0	OPEMODE[1:0]	Operating Mode Setting The access type from the EtherCAT master is R/W. 0 0: Buffer mode (3-buffer mode) 1 0: Mailbox mode (single buffer mode) Others: Reserved	R
3:2	DIR[1:0]	Transfer Direction Setting The access type from the EtherCAT master is R/W. 0 0: Read (ECAT: read access; PDI: write access) 0 1: Write (ECAT: write access; PDI: read access) Others: Reserved	R
4	IRQECAT	ECAT Event Interrupt Setting Enables or disables interrupts (ECAT interrupts) by the ECAT Event Request register (ECAT_EVENT_REQ at 0x0210). The access type from the EtherCAT master is R/W. 0: Disabled 1: Enabled	R
5	IRQPDI	AL Event Interrupt Setting Enables or disables interrupts (PDI interrupts) by the AL Event Request register (AL_EVENT_REQ at 0x0220). The access type from the EtherCAT master is R/W. 0: Disabled 1: Enabled	R
6	WDTRGEN	Watchdog Trigger Setting The access type from the EtherCAT master is R/W. 0: Disabled 1: Enabled	R
7	—	The read value is undefined.	R

Note: This register can only be written when SyncManager is disabled (bit 0 is 0 in the SyncManager Activate n register (SMn.ACT at 0x0806 + 0x8 × n)).

The SMn_CONTROL register is used to set operation of SyncManager.

36.16.4 SMn_STATUS : SyncManager Status n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0805 + 0x8 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	WRBU F	RDBU F	BUFFERED[1:0]		MAILB OX	—	INTRD	INTW R
Value after reset:	0	0	1	1	0	x	0	0

Bit	Symbol	Function	R/W
0	INTWR	Write Complete Interrupt State Indication 0: Indicates that the buffer was successfully written 1: Indicates that the first byte of the buffer was read (interrupt cleared)	R
1	INTRD	Read Complete Interrupt State Indication 0: Indicates that the buffer was successfully read 1: Indicates that the first byte of the buffer was written (interrupt cleared)	R
2	—	The read value is undefined.	R
3	MAILBOX	Mailbox Status Indication This bit is not used in buffer mode. 0: Mailbox empty 1: Mailbox full	R
5:4	BUFFERED[1:0]	Buffer Status Indication Indicates the buffer status in buffer mode (last written buffer). This bit is not used in mailbox mode. 0 0: 1st buffer 0 1: 2nd buffer 1 0: 3rd buffer 1 1: No buffer written	R
6	RDBUF	Read State Indication Indicates that the buffer is being read.	R
7	WRBUF	Write State Indication Indicates that the buffer is being written.	R

The SMn_STATUS register indicates the state of SyncManager.

36.16.5 SMn_ACT : SyncManager Activate n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0806 + 0x8 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	LATC HPDI	LATC HECA T	—	—	—	—	REPE ATRE Q	SMEN
Value after reset:	0	0	x	x	x	x	0	0

Bit	Symbol	Function	R/W
0	SMEN	SyncManager Enable/Disable The access type from the EtherCAT master is R/W. 0: Disabled. Memory is accessed without SyncManager control. 1: Enabled. SyncManager is active and controls memory area set in configuration.	R

Bit	Symbol	Function	R/W
1	REPEATREQ	Repeat Request Toggling of the repeat request signal means that retrying the mailbox is required (primarily used in conjunction with reading of the ECAT mailbox). The access type from the EtherCAT master is R/W.	R
5:2	—	The read values are undefined.	R
6	LATHECAT	ECAT Latch Event Specification The access type from the EtherCAT master is R/W. 0: No latch events 1: Generates latch events if the EtherCAT master switches the buffers	R
7	LATCHPDI	PDI Latch Event Specification The access type from the EtherCAT master is R/W. 0: No latch events 1: Generates latch events if the PDI switches the buffers or accesses the buffer start address	R

Note: Reading this register from the PDI in all SyncManagers which have changed activation clears bit 4 of the AL Event Request register (AL_EVENT_REQ at 0x0220).

The SMn_ACT register is used to set operation of SyncManager.

36.16.6 SMn_PDI_CONT : SyncManager PDI Control n Register (n = 0 to 7)

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0807 + 0x8 × n

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	REPEATACK	DEACTIVE

Value after reset: x x x x x x 0 0

Bit	Symbol	Function	R/W
0	DEACTIVE	SyncManager Operation Indication/Setting Deactivates SyncManager. Writing 1 is delayed until the end of a frame which is currently processed. The access type from the EtherCAT master is R. 0: Read: Normal operation. SyncManager is activated. Write: Activates SyncManager. 1: Read: SyncManager is deactivated and reset. SyncManager locks access to memory area. Write: Deactivates SyncManager.	R/W
1	REPEATACK	Repeat Acknowledge If this bit is set to the same value as set by bit 0 (repeat request) of the SyncManager activate register (SMn.ACT at 0x0806 + 0x8 × n), the PDI acknowledges the execution of a previous set repeat request. The access type from the EtherCAT master is R.	R/W
7:2	—	The read values are undefined. The write value should be 0.	R/W

The SMn_PDI_CONT register is used to set operation of SyncManager from the PDI.

C

RCVTIME0[31:0]

C

RCVTIME1[31:0]

C

RCVTIME2[31:0]

Bit	Symbol	Function	R/W
31:0	RCVTIME2[31:0]	Receive Time Indication Indicate the local time when port 2 starts to receive a frame (first start bit of preamble) containing the BWR or FPWR command to the receive time port 0 register (DC_RCV_TIME_PORT0 at 0x0900).	R

The DC_RCV_TIME_PORT2 register indicates the received time of the frame latched at port 2.

36.17.2 DC Time Loop Control Unit Registers

36.17.2.1 DC SYS TIME : System Time Register

Base address: ESC = 0x403A_0000
ESC NS = 0x503A_0000

Offset address: 0x0910

Bit position: 63 32

Bit field:	SYSTIME[63:32]
------------	----------------

Value after reset: 0

Bit position: 31 0

Bit field:	SYSTIME[31:0]
------------	---------------

Value after reset: 0

Bit	Symbol	Function	R/W
63:0	SYSTIME[63:0]	System Time Indication Access from the ECAT (read-write) Read: Indicate the local copy of the system time when the frame passed the reference clock (such as including a system time delay). The time latched at the start of frame (SOF) is indicated. Write: A written value is compared with the local copy of the system time. The result is input to the time control loop unit. Access from the PDI (read-only) Indicate the local copy of the system time. The time latched when the first byte of this register was read is indicated.	R

The DC SYS TIME register indicates the local copy of the system time.

36.17.2.2 DC RCV TIME UNIT : Receive Time ECAT Processing Unit Register

Base address: ESC = 0x403A_0000
ESC NS = 0x503A 0000

Offset address: 0x0918

Bit position: 63 32

Bit field:	RCVTIMEEPU[63:32]
------------	-------------------

[illegible]

Bit position: 31 0

Bit field:	RCVTIMEEPU[31:0]
------------	------------------

[illegible]

Bit	Symbol	Function	R/W
63:0	RCVTIMEEPU[63:0]	Receive Time Indication This register indicates the local time at the beginning of reception by the EtherCAT processing unit of a frame (such as the first start bit of the preamble), including write access to the Receive Time Port 0 register (DC_RCV_TIME_PORT0 at 0x0900).	R

Note: If port 0 is open, the value in this register reflects the same time as the value in the Receive Time Port 0 register (DC_RCV_TIME_PORT0 at 0x0900), but as 64 bits.

The DC_RCV_TIME_UNIT register indicates the received time of a frame latched at EtherCAT processing unit.

36.17.2.3 DC_SYS_TIME_OFFSET : System Time Offset Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0920

Bit position: 63

32

Bit field:

SYSTIMOFST[63:32]

Value after reset: 0

Bit position: 31

0

Bit field:

SYSTIMOFST[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
63:0	SYSTIMOFST[63:0]	System Time and Local Time Difference Indication Indicate a difference between the local time and system time. This offset is added to the local time to obtain the local system time. The access type from the EtherCAT master is R/W.	R

The DC_SYS_TIME_OFFSET register is used to indicate a difference (offset) between the local time and system time.

36.17.2.4 DC_SYS_TIME_DELAY : System Time Delay Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0928

Bit position: 31

0

Bit field:

SYSTIMDLY[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SYSTIMDLY[31:0]	Propagation Delay Indication Indicate a delay between the reference clock and the ESC. The access type from the EtherCAT master is R/W.	R

The DC_SYS_TIME_DELAY register indicates a propagation delay between the reference clock and slave (ESC).

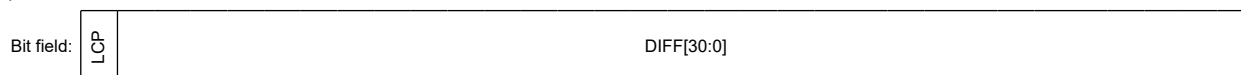
36.17.2.5 DC_SYS_TIME_DIFF : System Time Difference Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x092C

Bit position: 31 30

0



Value after reset: 0

Bit	Symbol	Function	R/W
30:0	DIFF[30:0]	System Time Mean Difference Indication Indicates a mean difference between the local copy of the system time and received system time.	R
31	LCP	System Time Greater/Less Indication Indicates whether the local copy of the system time is greater than or equal to, or is less than, the latest received copy of the system time. 0: Local copy of the system time greater than or equal to the received system time 1: Local copy of the system time less than the received system time	R

The DC_SYS_TIME_DIFF register indicates a mean difference between the local copy of the system time and received system time.

36.17.2.6 DC_SPEED_COUNT_START : Speed Counter Start Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0930

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0



Value after reset: x 0 0 1 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
14:0	SPDCNTSTRT[14:0]	Drift Correction Bandwidth Setting Indicate the bandwidth for adjustment of the local copy of the system time (larger values → smaller bandwidth and smoother adjustment). A write access resets the System Time Difference register (DC_SYS_TIME_DIFF at 0x092C) and the Speed Counter Difference register (DC_SPEED_COUNT_DIFF at 0x0932). Valid range: 0x0080 to 0x3FFF The access type from the EtherCAT master is R/W.	R
15	—	The read value is undefined. The access type from the EtherCAT master is R/W. In this case, the write value should be 0.	R

The DC_SPEED_COUNT_START register is used to set the bandwidth for drift correction of the local copy of the system time.

36.17.2.7 DC_SPEED_COUNT_DIFF : Speed Counter Difference Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0932

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: SPDCNTDIFF[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	SPDCNTDIFF[15:0]	Clock Period Deviation Indication Indicate the deviation between the local clock period and the clock period of the reference clock (represented by two's complements). Range: $\pm$ (speed counter start value - 0x7F)	R

The DC_SPEED_COUNT_DIFF register indicates the deviation between the local clock period and the clock period of the reference clock.

36.17.2.8 DC_SYS_TIME_DIFF_FIL_DEPTH : System Time Difference Filter Depth Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0934

Bit position: 7 6 5 4 3 2 1 0

Bit field: — — — — SYSTMDEP[3:0]

Value after reset: x x x x 0 1 0 0

Bit	Symbol	Function	R/W
3:0	SYSTMDEP[3:0]	Filter Depth Setting Set the filter depth for averaging the received system time deviation. A write access resets the System Time Difference register (DC_SYS_TIME_DIFF at 0x092C). The access type from the EtherCAT master is R/W.	R
7:4	—	The read values are undefined. The access type from the EtherCAT master is R/W. In this case, the write value should be 0.	R

The DC_SYS_TIME_DIFF_FIL_DEPTH register is used to set the filter depth for averaging the received system time deviation.

36.17.2.9 DC_SPEED_COUNT_FIL_DEPTH : Speed Counter Filter Depth Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0935

Bit position: 7 6 5 4 3 2 1 0

Bit field: — — — — CLKPERDEP[3:0]

Value after reset: x x x x 1 1 0 0

Bit	Symbol	Function	R/W
3:0	CLKPERDEP[3:0]	Filter Depth Setting Set the filter depth for averaging the clock period deviation. A write access resets the internal speed counter filter. The access type from the EtherCAT master is R/W.	R
7:4	—	The read values are undefined. The access type from the EtherCAT master is R/W. In this case, the write value should be 0.	R

The DC_SPEED_COUNT_FIL_DEPTH register is used to set the filter depth for averaging the clock period deviation.

36.17.3 Cyclic Unit Control Registers

36.17.3.1 DC_CYC_CONT : Cyclic Unit Control Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0980

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	LATC H1	LATC H0	—	—	—	SYNC OUT

Value after reset: x x 0 0 x x x 0

Bit	Symbol	Function	R/W
0	SYNCOUT	SYNC Output Unit Control Setting The access type from the EtherCAT master is R/W. 0: ECAT control 1: PDI control	R
3:1	—	The read values are undefined.	R
4	LATCH0	Latch Input Unit 0 Control Setting Latch interrupt is routed to the ECAT or PDI in accordance with this setting. The access type from the EtherCAT master is R/W. 0: ECAT control 1: PDI control	R
5	LATCH1	Latch Input Unit 1 Control Setting Latch interrupt is routed to the ECAT or PDI in accordance with this setting. The access type from the EtherCAT master is R/W. 0: ECAT control 1: PDI control	R
7:6	—	The read values are undefined.	R

The DC_CYC_CONT register sets whether to control SYNC and latch units by the ECAT or PDI.

36.17.4 SYNC Output Unit Registers

36.17.4.1 DC_ACT : Activation Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0981

Bit position:	7	6	5	4	3	2	1	0
Bit field:	DBGP ULSE	NEAR FUTU RE	START TIME	EXTS TARTT IME	AUTO ACT	SYNC 1	SYNC 0	SYNC ACT

Value after reset: 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
0	SYNCACT	Sync Output Unit Activation Activates the Sync output unit. Write 1 after the start time is written. 0: Deactivated 1: Activated	R/W
1	SYNC0	SYNC0 Output Setting Sets SYNC0 output. 0: Deactivated 1: SYNC0 pulse output is generated	R/W
2	SYNC1	SYNC1 Output Setting Sets SYNC1 output. 0: Deactivated 1: SYNC1 pulse output is generated	R/W
3	AUTOACT	SYNC Output Unit Activation Sets whether to activate the Sync output unit automatically by writing to the Start Time Cyclic Operation register (DC_CYC_START_TIME at 0x0990): 0: Deactivated 1: Activated. Bit 0 is automatically set to 1 in this register after the start time is written.	R/W
4	EXTSTARTTIME	Start Time Cyclic Operation Extension Extends start time cyclic operation. 0: No extension 1: Extends the start time written with 32 bits to 64 bits	R/W
5	STARTTIME	Start Time Plausibility Selects whether checking the plausibility of the start time and response to implausible start times is to proceed. 0: Disabled. Sync signal is generated if the start time is reached. 1: Sync signal is generated immediately if the start time is outside the range of the near future.	R/W
6	NEARFUTURE	Near Future Range Setting Sets the range to be considered the near future. 0: Up to 2^{63} ns from now (1/2 of the DC width) 1: Up to 2^{31} ns from now (about 2.1 sec.)	R/W
7	DBGPULSE	Debug Pulse Setting Sets Sync signal debug pulse. This bit is self-cleared and always read as 0. 0: Deactivated 1: Immediately generates a single debug ping on the SYNC0 and SYNC1 pins in accord with the setting of bits 2 and 1 of this register.	R/W

Note: Writing to this register depends on the setting of bit 0 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_ACT register is used to activate the Sync output unit.

36.17.4.2 DC_PULSE_LEN : SYNC Signal Pulse Length Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0982

Bit position: 15 14 13 12 11 10 9 8 7 6 5 4 3 2 1 0

Bit field: PULSELEN[15:0]

Value after reset: 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0 0

Bit	Symbol	Function	R/W
15:0	PULSELEN[15:0]	SYNC Signal Pulse Length Indication Indicate the pulse length of SYNC signals (in units of 10 ns) 0: Acknowledge mode. In this mode, SYNC signal is cleared by reading the SYNC0 or SYNC1 status register (DC_SYNC0/1_STAT at 0x098E, 0x098F).	R

The DC_PULSE_LEN register indicates the pulse length of SYNC signals.

36.17.4.3 DC_ACT_STAT : Activation Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0984

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	START TIME	SYNC 1ACT	SYNC 0ACT
Value after reset:	x	x	x	x	x	0	0	0

Bit	Symbol	Function	R/W
0	SYNC0ACT	SYNC0 Status Indication Indicates the activation state of SYNC0. 0: First SYNC0 pulse is not pending 1: First SYNC0 pulse is pending	R
1	SYNC1ACT	SYNC1 Status Indication Indicates the activation state of SYNC1. 0: First SYNC1 pulse is not pending 1: First SYNC1 pulse is pending	R
2	STARTTIME	Plausibility Result Indication Indicates the plausibility check result of the Start Time Cyclic Operation register (DC_CYC_START_TIME at 0x0990) while the Sync output unit is activated. 0: The start time is within the near future 1: The start time is out of the near future	R
7:3	—	The read values are undefined.	R

The DC_ACT_STAT register indicates the activation status of SYNC output signals.

36.17.4.4 DC_SYNC0_STAT : SYNC0 Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x098E

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SYNC 0STA
Value after reset:	x	x	x	x	x	x	x	0

Bit	Symbol	Function	R/W
0	SYNC0STA	SYNC0 State Indication Indicates the SYNC0 state for acknowledge mode. SYNC0 in acknowledge mode is cleared by reading this register from the PDI. This bit is only used in acknowledge mode.	R
7:1	—	The read values are undefined.	R

The DC_SYNC0_STAT register indicates the state of SYNC0 output. It is only used in acknowledge mode.

36.17.4.5 DC_SYNC1_STAT : SYNC1 Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x098F

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	—	SYNC1STA

Value after reset: x x x x x x x 0

Bit	Symbol	Function	R/W
0	SYNC1STA	SYNC1 State Indication Indicates the SYNC1 state for acknowledge mode. SYNC1 in acknowledge mode is cleared by reading this register from the PDI. This bit is only used in acknowledge mode.	R
7:1	—	The read values are undefined.	R

The DC_SYNC1_STAT register indicates the state of SYNC1 output. It is only used in acknowledge mode.

36.17.4.6 DC_CYC_START_TIME : Start Time Cyclic Operation/Next SYNC0 Pulse Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0990

Bit position:	63	32
Bit field:	STATIM[63:32]	

Value after reset: 0

Bit position:	31	0
Bit field:	STATIM[31:0]	

Value after reset: 0

Bit	Symbol	Function	R/W
63:0	STATIM[63:0]	Start Time Setting/System Time Indication Write: Set the start time (in the system time) of cyclic operation in ns units. Read: Indicate the system time of the next SYNC0 pulse in ns units.	R/W

Note:

- Writing to this register depends on the setting of bit 0 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980). Only writable when bit 0 is 0 in the SYNC Activation register (DC_ACT at 0x0981).
- When auto-activation is enabled, the upper 32 bits are automatically extended if only the lower 32 bits are written within one frame.

Writing to this register sets the start time of cyclic operation. Reading this register indicates the system time of the next SYNC0 pulse.

36.17.4.7 DC_NEXT_SYNC1_PULSE : Next SYNC1 Pulse Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x0998

Bit position: 63

32

Bit field:

SYNC1PULSE[63:32]

Value after reset: 0

Bit position: 31

0

Bit field:

SYNC1PULSE[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
63:0	SYNC1PULSE[63:0]	SYNC1 Pulse System Time Indication Indicate the system time of the next SYNC1 pulse in ns units.	R

The DC_NEXT_SYNC1_PULSE register indicates the system time of the next SYNC1 pulse.

36.17.4.8 DC_SYNC0_CYC_TIME : SYNC0 Cycle Time Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09A0

Bit position: 31

0

Bit field:

SYNC0CYC[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SYNC0CYC[31:0]	Time Between Consecutive SYNC0 Pulses Set the time between two consecutive SYNC0 pulses in ns units. 0: Single-shot mode. Only one SYNC0 pulse is generated in single-shot mode.	R/W

Note: Writing to this register depends on the setting of bit 0 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_SYNC0_CYC_TIME register is used to set the time between consecutive SYNC0 pulses.

36.17.4.9 DC_SYNC1_CYC_TIME : SYNC1 Cycle Time Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09A4

Bit position: 31

0

Bit field:

SYNC1CYC[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	SYNC1CYC[31:0]	Time between SYNC1 and SYNC0 Pulses Set the time between SYNC1 and SYNC0 pulses in ns units.	R/W

Note: Writing to this register depends on the setting of bit 0 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_SYNC1_CYC_TIME register is used to set the time between SYNC1 and SYNC0 pulses.

36.17.5 Latch Input Unit Registers

36.17.5.1 DC_LATCH0_CONT : Latch 0 Control Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09A8

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	NEGE DGE	POSE DGE

Value after reset: x x x x x x 0 0

Bit	Symbol	Function	R/W
0	POSEDGE	Latch 0 Positive Edge Function Setting Sets the function of the rising edge of the latch 0 input signal. 0: Continuous latch active 1: Single event (only first event active)	R/W
1	NEGEDGE	Latch 0 Negative Edge Function Setting Sets the function of the falling edge of the latch 0 input signal. 0: Continuous latch active 1: Single event (only first event active)	R/W
7:2	—	The read values are undefined. The write value should be 0. The access type from the EtherCAT master is R.	R/W

Note: Writing to this register depends on the setting of bit 4 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_LATCH0_CONT register is used to control the edge function of the latch 0 input signal.

36.17.5.2 DC_LATCH1_CONT : Latch 1 Control Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09A9

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	—	NEGE DGE	POSE DGE

Value after reset: x x x x x x 0 0

Bit	Symbol	Function	R/W
0	POSEDGE	Latch 1 Positive Edge Function Setting Sets the function of the rising edge of the latch 1 input signal. 0: Continuous latch active 1: Single event (only first event active)	R/W
1	NEGEDGE	Latch 1 Negative Edge Function Setting Sets the function of the falling edge of the latch 1 input signal. 0: Continuous Latch active 1: Single event (only first event active)	R/W
7:2	—	The read values are undefined. The write value should be 0. The access type from the EtherCAT master is R.	R/W

Note: Writing to this register depends on the setting of bit 5 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_LATCH1_CONT register is used to control the edge function of the latch 1 input signal.

36.17.5.3 DC_LATCH0_STAT : Latch 0 Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09AE

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	PINST ATE	EVEN TNEG	EVEN TPOS

Value after reset: x x x x x 0 0 0

Bit	Symbol	Function	R/W
0	EVENTPOS	Latch 0 Positive Edge Event Indication Indicates detection of rising edges of the event latch 0 signal. This flag is cleared by reading the Latch 0 Time Positive Edge register (DC_LATCH0_TIME_POS at 0x09B0). 0: Rising edge not detected or continuous mode 1: Rising edge detected and mode is single-event	R
1	EVENTNEG	Latch 0 Negative Edge Event Indication Indicates detection of falling edges of the event latch 0 signal. This flag is cleared by reading the Latch 0 Time Negative Edge register (DC_LATCH0_TIME_NEG at 0x09B8). 0: Falling edge not detected or continuous mode 1: Falling edge detected and mode is single-event	R
2	PINSTATE	Latch 0 Input Pin State Indication Indicates the state of the latch 0 input pin.	R
7:3	—	The read values are undefined.	R

The DC_LATCH0_STAT register indicates the state of the latch 0 input signal.

36.17.5.4 DC_LATCH1_STAT : Latch 1 Status Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09AF

Bit position:	7	6	5	4	3	2	1	0
Bit field:	—	—	—	—	—	PINST ATE	EVEN TNEG	EVEN TPOS

Value after reset: x x x x x 0 0 0

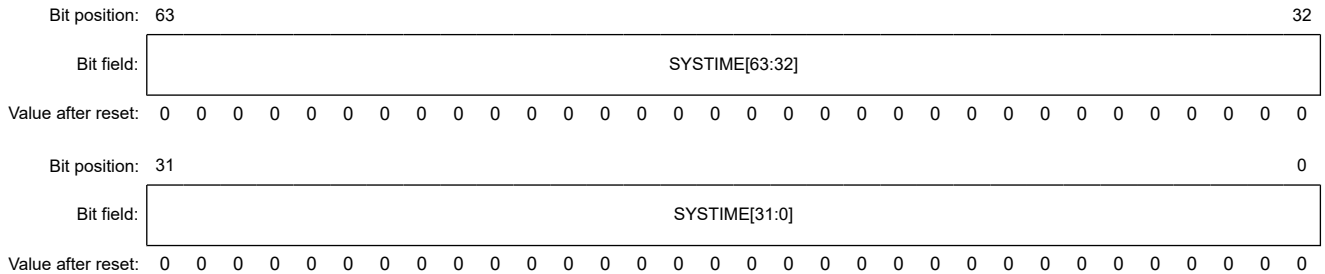
Bit	Symbol	Function	R/W
0	EVENTPOS	Latch 1 Positive Edge Event Indication Indicates detection of rising edges of the event latch 1 signal. This flag is cleared by reading the Latch 1 Time Positive Edge register (DC_LATCH1_TIME_POS at 0x09C0). 0: Rising edge not detected or continuous mode 1: Rising edge detected and mode is single-event	R
1	EVENTNEG	Latch 1 Negative Edge Event Indication Indicates detection of falling edges of the event latch 1 signal. This flag is cleared by reading the Latch 1 Time Negative Edge register (DC_LATCH1_TIME_NEG at 0x09C8). 0: Falling edge not detected or continuous mode 1: Falling edge detected and mode is single-event	R
2	PINSTATE	Latch 1 Input Pin State Indication Indicates the state of the latch 1 input pin.	R
7:3	—	The read values are undefined.	R

The DC_LATCH1_STAT register indicates the state of the latch 1 input signal.

36.17.5.5 DC_LATCH0_TIME_POS : Latch 0 Time Positive Edge Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09B0



Bit	Symbol	Function	R/W
63:0	SYSTIME[63:0]	System Time Indication Indicate the system time captured at the rising edge of the latch 0 input signal. Reading this register clears bit 0 of the Latch 0 Status register (DC_LATCH0_STAT at 0x09AE).	R

Note:

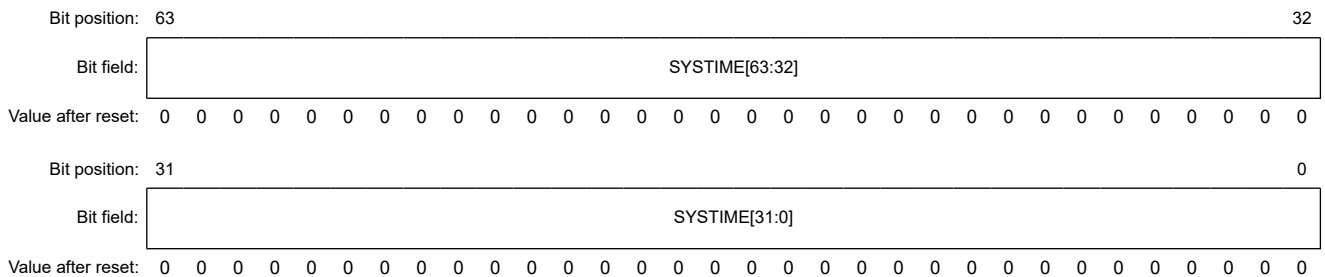
- Bits [63:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value
- Clearing the latch 0 status flag function depends on the setting of bit 4 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_LATCH0_TIME_POS register indicates the system time at the rising edge of the latch 0 input signal.

36.17.5.6 DC_LATCH0_TIME_NEG : Latch 0 Time Negative Edge Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09B8



Bit	Symbol	Function	R/W
63:0	SYSTIME[63:0]	System Time Indication Indicate the system time captured at the falling edge of the latch 0 input signal. Reading this register clears bit 1 of the Latch 0 Status register (DC_LATCH0_STAT at 0x09AE).	R

Note:

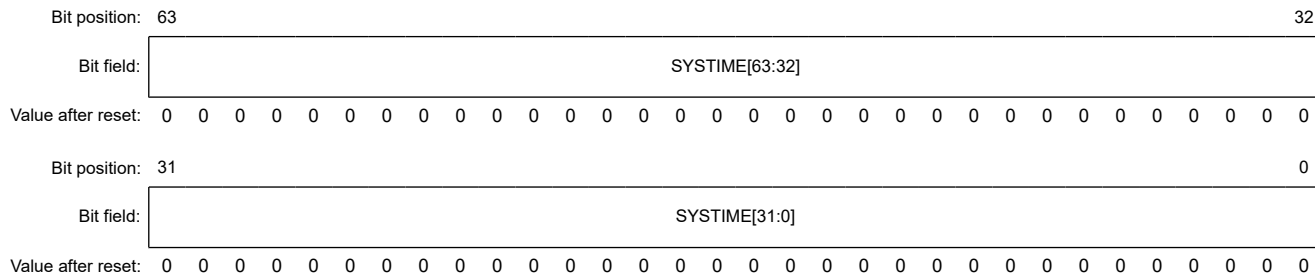
- Bits [63:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value
- Clearing the latch 0 status flag function depends on the setting of bit 4 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_LATCH0_TIME_NEG register indicates the system time at the falling edge of the latch 0 input signal.

36.17.5.7 DC_LATCH1_TIME_POS : Latch 1 Time Positive Edge Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09C0



Bit	Symbol	Function	R/W
63:0	SYSTIME[63:0]	System Time Indication Indicate the system time captured at the rising edge of the latch 1 input signal. Reading this register clears bit 0 of the Latch 1 Status register (DC_LATCH1_STAT at 0x09AF).	R

Note:

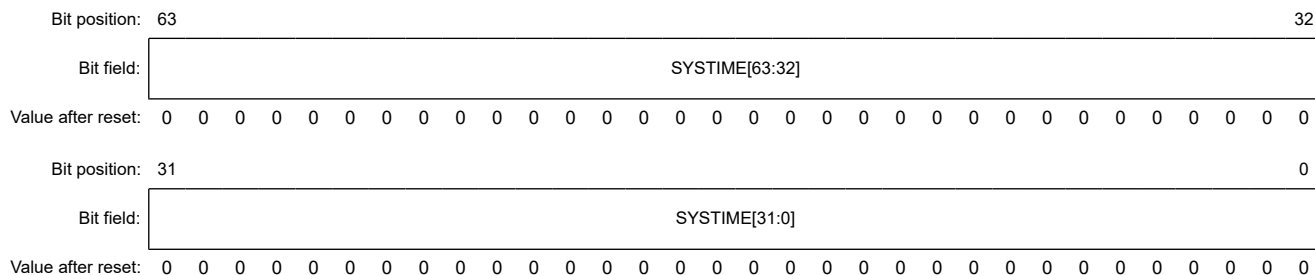
- Bits [63:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value
- Clearing the latch 1 status flag function depends on the setting of bit 5 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_LATCH1_TIME_POS register indicates the system time at the rising edge of the latch 1 input signal.

36.17.5.8 DC_LATCH1_TIME_NEG : Latch 1 Time Negative Edge Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09C8



Bit	Symbol	Function	R/W
63:0	SYSTIME[63:0]	System Time Indication Indicate the system time captured at the falling edge of the latch 1 input signal. Reading this register clears bit 1 of the Latch 1 Status register (DC_LATCH1_STAT at 0x09AF).	R

Note:

- Bits [63:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value
- Clearing the latch 1 status flag function depends on the setting of bit 5 of the Cyclic Unit Control register (DC_CYC_CONT at 0x0980).

The DC_LATCH1_TIME_NEG register indicates the system time at the falling edge of the latch 1 input signal.

36.17.6 SyncManager Event Time Registers

36.17.6.1 DC_EC_CNG_EV_TIME : Buffer Change Event Time Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09F0

Bit position: 31 0

Bit field: ECATCHANGE[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	ECATCHANGE[31:0]	Local Time Indication Indicate the local time at the beginning of a frame which causes at least one SyncManager to generate an ECAT event (switching the buffers).	R

Note: Bits [31:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value.

The DC_EC_CNG_EV_TIME register indicates the local time at the beginning of a frame which causes SyncManager to generate an ECAT event (switching the buffers).

36.17.6.2 DC_PDI_START_EV_TIME : PDI Buffer Start Event Time Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09F8

Bit position: 31 0

Bit field: PDISTART[31:0]

Value after reset: 0

Bit	Symbol	Function	R/W
31:0	PDISTART[31:0]	Local Time Indication Indicate the local time when at least one SyncManager has generated a PDI event (access to the address where a buffer starts).	R

Note: Bits [31:8] are internally latched (ECAT/PDI independently) when bits [7:0] are read, which guarantees reading a consistent value.

The DC_PDI_START_EV_TIME register indicates the local time when SyncManager has generated a PDI event (access to the address where a buffer starts).

36.17.6.3 DC_PDI_CNG_EV_TIME : PDI Buffer Change Event Time Register

Base address: ESC = 0x403A_0000
ESC_NS = 0x503A_0000

Offset address: 0x09FC

Bit position: 31 0

Bit field: PDICHANGE[31:0]

Value after reset: 0

0x503A_0F80 to 0x503A_0FFF. An initial value of 1 indicates that the implementation of this module supports the corresponding feature, except in the case of bits [7:0], which indicate the number of bits defined in the user RAM and the initial value is 0x33 for this module.

Bit position	Description	Initial value
7:0	Number of bits for defining extended functionality. The value is 51 (0x33) for this module.	0x33
8	Extended DL Control register (0x0102, 0x0103)	1
9	AL Status Code register (0x0134, 0x0135)	1
10	ECAT event mask (0x0200, 0x0201)	1
11	Configured station alias (0x0012, 0x0013)	1
12	General input (0x0F18, 0x0F1F)	0
13	General output (0x0F10, 0x0F17)	0
14	AL event mask (0x0204, 0x0207)	1
15	Physical read/write offset (0x0108, 0x0109)	1
16	Watchdog divider writable (0x0400, 0x0401) and watchdog PDI (0x0410, 0x0F11)	1
17	Watchdog counter (0x0442, 0x0443)	1
18	Write protection (0x0020, 0x0031)	1
19	Reset (0x0040, 0x0041)	1
20	Reserved	0
21	DC SyncManager event time (0x09F0, 0x09FF)	1
22	ECAT processing unit/PDI error counter (0x030C, 0x030D)	1
23	EEPROM size configurable (bit 7 at 0x0502) 0: EEPROM size fixed up to 16 Kbits 1: EEPROM size configurable	1
26:24	Reserved	0
27	Lost link counter (0x0310, 0x0313)	1
28	MII management interface (0x0510, 0x0515)	1
29	Enhanced link detection MII	1
30	Enhanced link detection EBUS	0
31	RUN LED	1
32	Link/activity LED	1
33	Reserved	0
35:34	Reserved	1
36	Reserved	0
37	Reserved	1
38	DC Time loop control assigned to PDI	0
39	Link detection and configuration by MI	0
40	MI control by PDI	1
41	Automatic TX shift	1
42	EEPROM emulation	0
49:43	Reserved	0
50	ERR LED, RUN/ERR LED override	1
Others	Reserved	Reserved

36.18.4 DATA_RAM : Process Data RAM

The process data RAM is used for processing data and mailbox and takes up 8 KB from 0x403A_1000 to 0x403A_2FFF in secure region address accesses. In non-secure region address access, it is from 0x503A_1000 to 0x503A_2FFF.

This RAM is only accessible when the EEPROM is correctly loaded (for example, when bit 0 is 1 in the ESC DL Status register, ESC_DL_STATUS, at 0x0110).

36.19 Operation

36.19.1 Clocks

The ESC uses the ESCCLK which is generated by the clock generation circuit and the ESCCLK divided by 4 as CLK100 and CLK25 of the ESC IP Core, respectively. Set the registers in the clock generation circuit so that the frequency of the ESCCLK becomes 100 MHz.

Also, set the registers so that the frequency of the ETHPHYCLK signal which is used as the operating clock of a PHY-LSI becomes 25 MHz.

36.19.2 Initial Settings

Follow the procedure below to initialize the ESC.

1. Release the ESC from the module stop state by using the MSTPCRC.MSTPC28 bit and MSTPCRC.MSTPC29 bit.
2. Set the amount of offset for the PHY addresses in the ECATOFFADR register.
3. Set the size of the EEPROM in the ECATOPMOD register.
4. Set the additional delays for manual TX shift compensation in the ECATDBGC register.
5. Release the ESC IP Core and the PHY-LSI from the reset state by using the ESCRST and PHYRST bits in the ESCRST register.

On release from the reset state, the ESC will automatically load the EEPROM data and be activated.

36.19.3 Configuration of the Reset Circuit

Figure 36.3 shows the configuration of the reset circuit of the ESC. On reception of a reset request from the ECAT (ESC_RESET_ECAT register) or a reset request from the PDI (ESC_RESET_PDI register), the ESC IP Core stops and the reset output from the ESC IP Core becomes high. The reset output from the ESC IP Core drives the signal on the CATRESTOUT pin low and the externally connected PHY-LSI is reset. A RESET interrupt is generated at the same time.

To release the ESC IP Core from the reset state, the ESCRST.ESCRST bit must be switched to 0 and then to 1 after the RESET interrupt is detected. Note that the reset output from the ESC IP Core becomes 0 when the reset input to the ESC IP Core changes from 1 to 0. Restarting of the ESC IP Core begins when the reset input to the ESC IP Core changes from 0 to 1 and loading of the EEPROM starts. Loading of the EEPROM is completed in about 1 ms. The timing with which the PHY-LSI is released from the reset state must be set so that it will only recommence operations after the ESC IP Core has started. Figure 36.4 shows the timing chart. Resetting the ESC IP Core by using the ESCRST.ESCRST bit instead of by the ECAT or PDI issuing a reset request is also possible. In this case, the CATRESTOUT pin does not automatically become low, so use the ESCRST.PHYRST bit to place the PHY-LSI in the reset state beforehand. Figure 36.5 shows the timing chart.

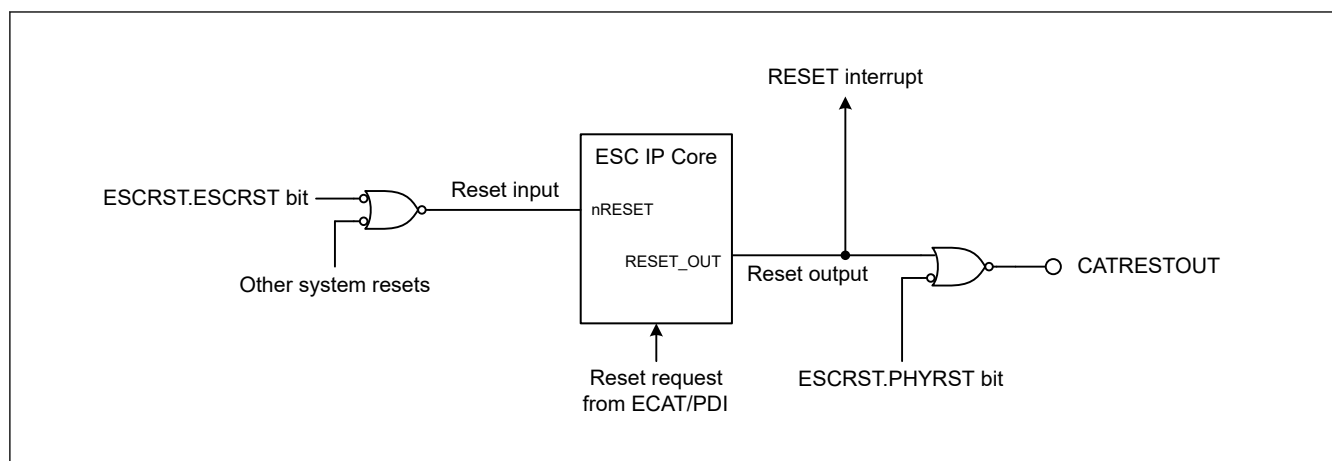


Figure 36.3 Configuration of the Reset Circuit of the ESC

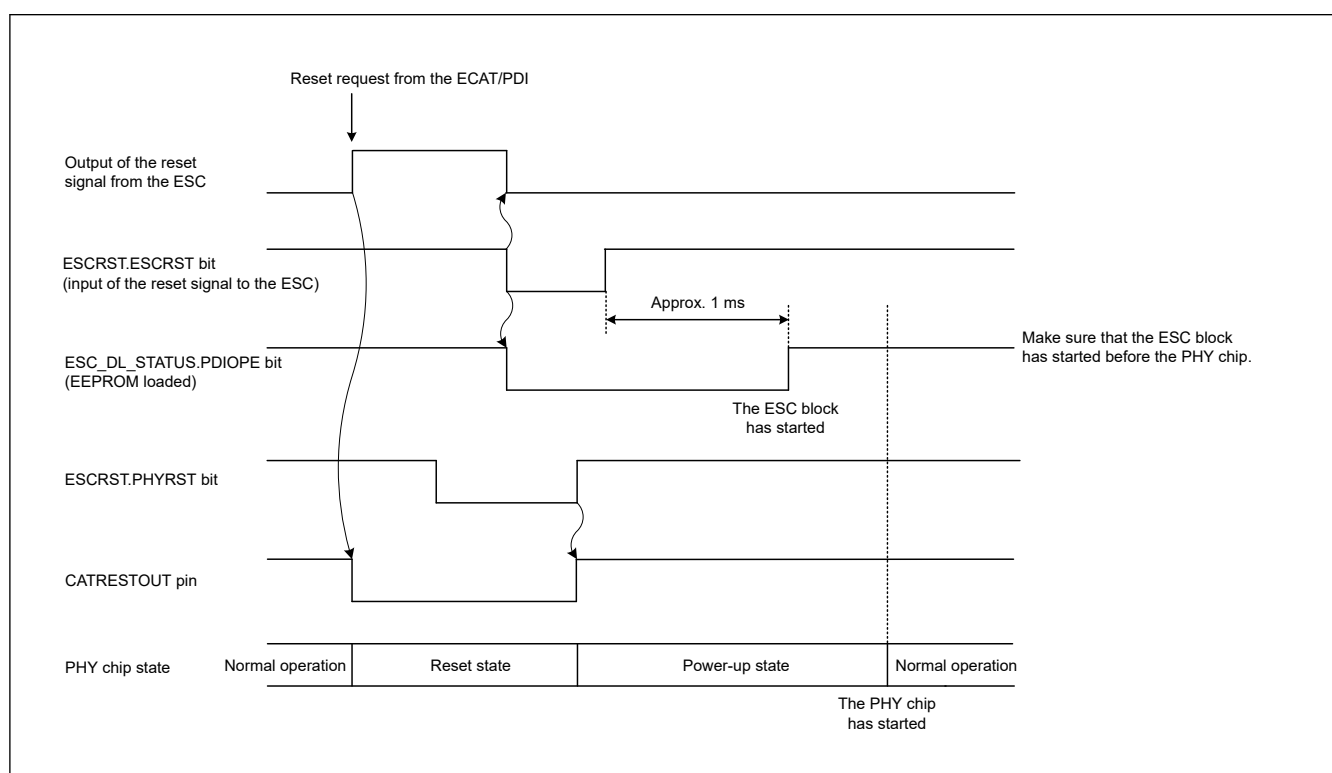


Figure 36.4 ESC Reset Timing (When a Reset is Requested by the ECAT or PDI)

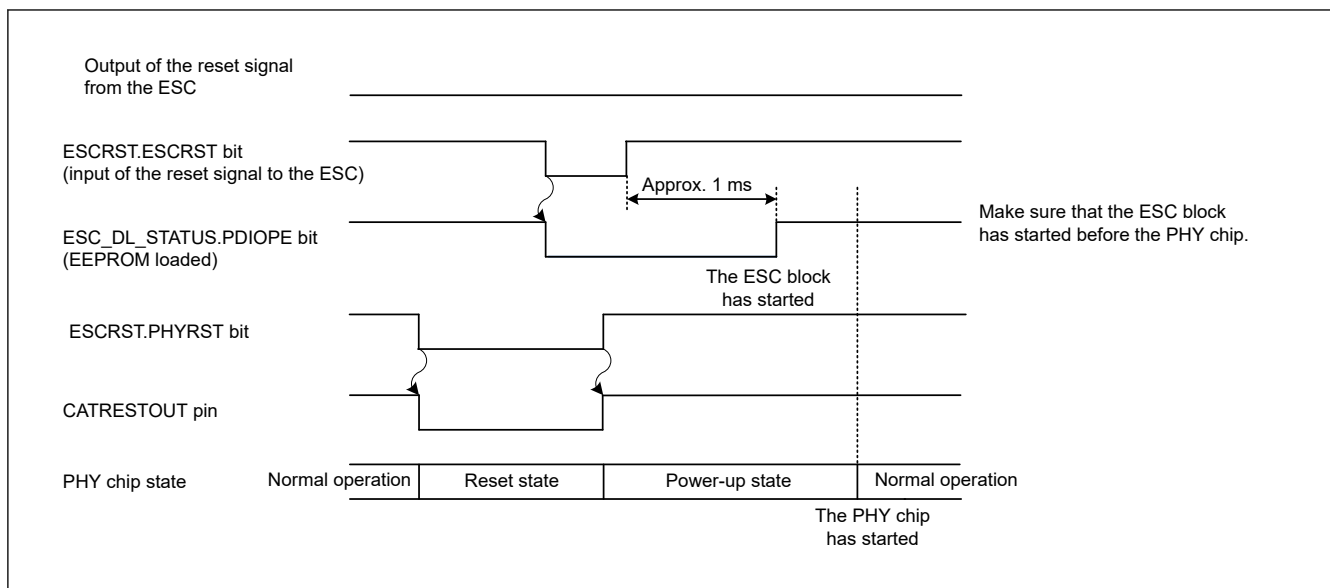


Figure 36.5 ESC Reset Timing (When the ESCRST.ESCRST Bit is Used to Reset the ESC)

36.19.4 Interrupts

Table 36.4 lists the interrupt requests generated by the ESC.

Table 36.4 ESC Interrupt Requests

Name	Description	Generating Condition
SYNC0	Sync0 interrupt	Rising or falling edges of the SYNC0 signal ^{*1}
SYNC1	Sync1 interrupt	Rising or falling edges of the SYNC1 signal ^{*1}
ESCI	EtherCAT interrupt	The PDI_IRQ signal is 1 or 0 ^{*1}
SOF	Start-of-Frame interrupt	The PDI_SOF signal changing from 0 to 1 ^{*2}
EOF	End-of-Frame interrupt	The PDI_EOF signal changing from 0 to 1 ^{*2}
WDT	Process Data Watchdog interrupt	The PDI_WD_STATE signal changing from 1 to 0
RESET	RESET interrupt	The RESET_OUT signal changing from 0 to 1

Note 1. The condition for generation can be selected by the ESCICR register

Note 2. When using SOF and EOF interrupts, set the PCLKA frequency to 37.5MHz or higher.

36.19.5 Event link

Table 36.5 lists the interrupt requests generated by the ESC.

Table 36.5 ESC event input

Name	Description	Generating Condition
ESC_LATCH0	EtherCAT LATCH0 event	ELC_ESC0 event assert
ESC_LATCH1	EtherCAT LATCH1 event	ELC_ESC1 event assert

Note: ESC_LATCHn event can be selected by ESCICR.LATnS bit (n=0,1).

Note: ESC_LATCHn event is only valid for rising edges.

36.19.6 Usage Notes

36.19.6.1 Setting the Module-Stop Function

Operation of the ESC can be enabled or disabled by setting a MSTPCRC.MSTPC29 bit.

The ESC is initially disabled after a reset. Registers in the ESC only become accessible after it has been released from the module stop state. For details, refer to section 11, Low Power Consumption.

Before placing the ESC in the module stop state, set the ESCRST.ESCRST bit to 0. If a transition to the module stop state is made with the ESCRST.ESCRST bit set to 1, ESC operation after exit from the module stop state cannot be guaranteed.

36.19.6.2 Software Standby Mode

Before transitioning to Software Standby mode, set the ESCRST.ESCRST bit to 0.

After returning from Software Standby mode, reinitialize the ESC.